ECO2011 Basic Microeconomics

Review Slides (Part I – Part V)

2026 Spring

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Competitive Market

- Ten principles
- Opportunity Cost
  - Whatever must be given up to obtain some item

Competitive Market

- Ten principles
  - Principle 1: People Face Trade-offs
  - Principle 2: The Cost of Something Is What You Give Up to Get It
  - Principle 3: Rational People Think at the Margin
  - Principle 4: People Respond to Incentives
  - Principle 5: Trade Can Make Everyone Better Off
  - Principle 6: Markets Are Usually a Good Way to Organize Economic Activity

Competitive Market

- Ten principles
  - Principle 7: Governments Can Sometimes Improve Market Outcomes
  - Principle 8: Country's Standard of Living Depends on Its Ability to Produce Goods and Services
  - Principle 9: Prices Rise When the Government Prints Too Much Money
  - Principle 10: Society Faces a Short-run Trade-off between Inflation and Unemployment

Comparative Advantage

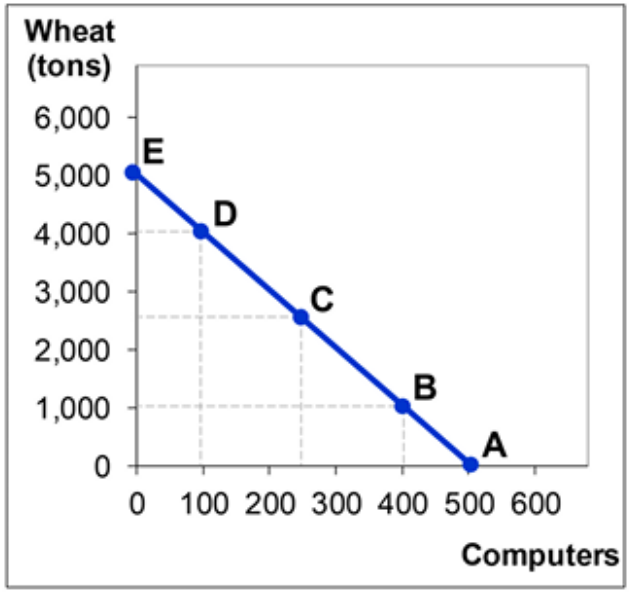
- Principle 5: Trade Can Make Everyone Better Off → Interdependence
- Production Possibilities Frontier (PPF)
- Comparative Advantage vs. Absolute Advantage

Comparative Advantage

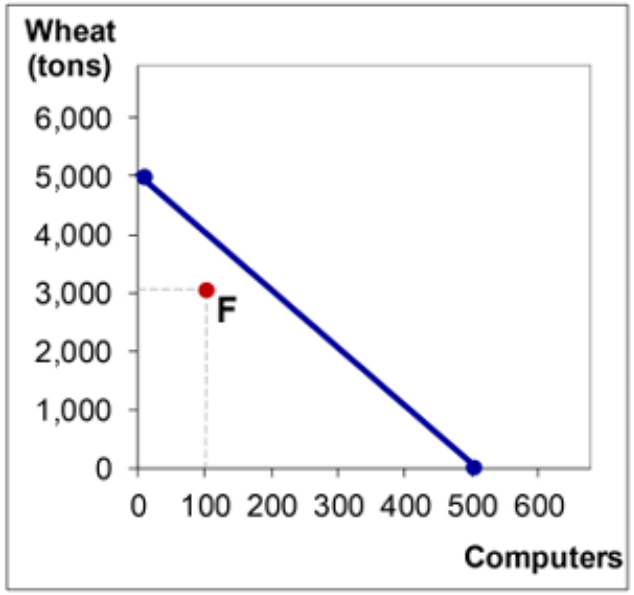
- Production Possibilities Frontier (PPF):
  - A graph that shows the **combinations of two goods** the economy can possibly produce, **given the available resources and the available technology**.

Comparative Advantage

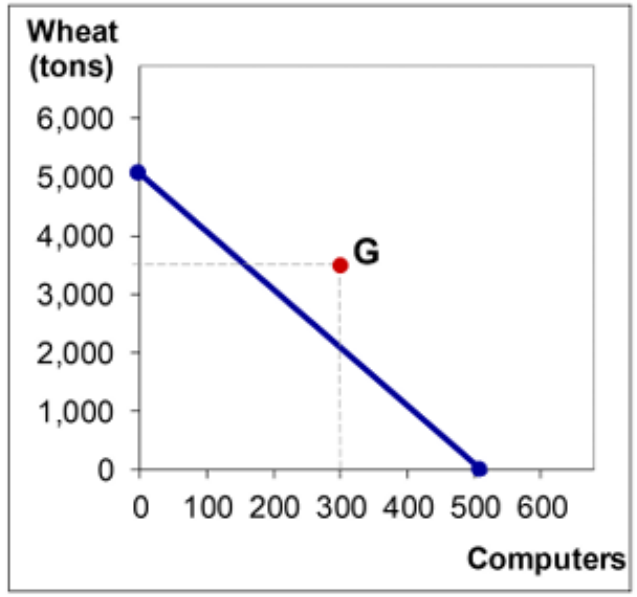
- Production Possibilities Frontier (PPF)



Possible/attainable and efficient



Possible but not efficient



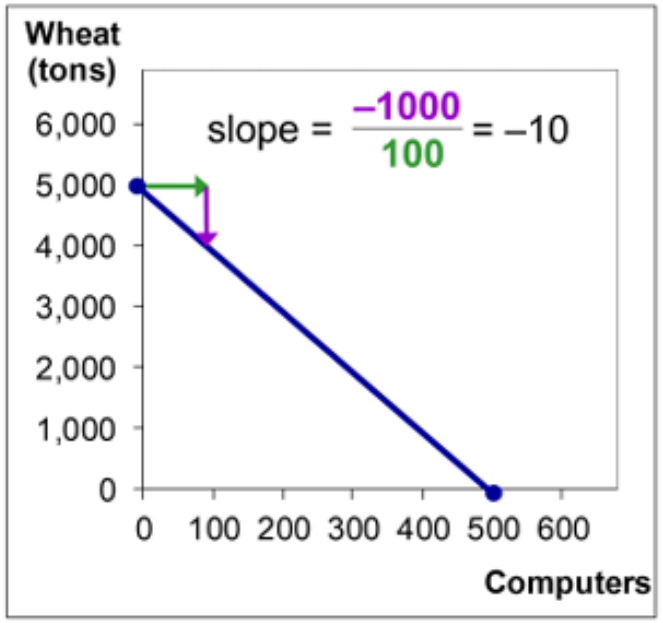
Not possible (limited resources)

Comparative Advantage

- Production Possibilities Frontier (PPF)

The **slope** of the PPF:

Opportunity cost of one good (x) in terms of the other (y).



Example:

Opportunity cost of 1 computer = 10 tons of wheat

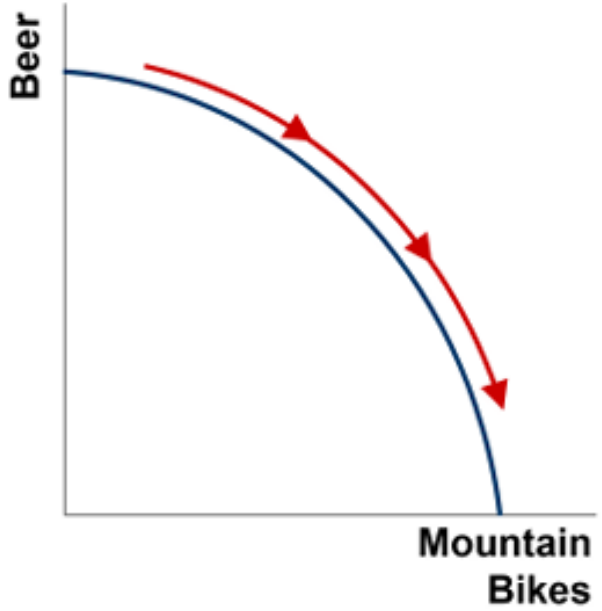
Comparative Advantage

- Production Possibilities Frontier (PPF)

The **shape** of the PPF:

PPF becomes steeper, and the opportunity cost of mountain bikes (x) increases

WHY?

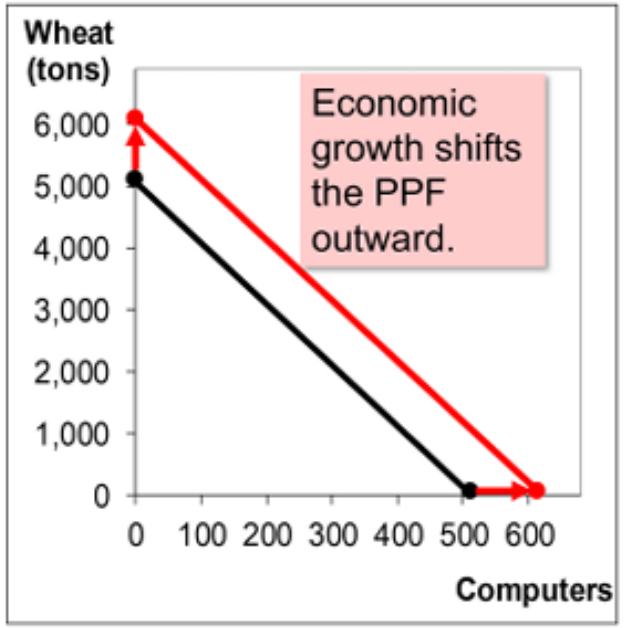


Comparative Advantage

- Production Possibilities Frontier (PPF)

The **shift** of the PPF:

Additional **resources** or an improvement in **technology**



Comparative Advantage

- Absolute Advantage:
  - The ability to produce a good using **fewer** inputs than another producer
  - Measures the cost of a good in terms of the inputs required to produce it
- Comparative Advantage:
  - The ability to produce a good at a **lower opportunity cost** than another producer
  - **Flatter PPF** → Comparative Advantage in x axis goods

Demand and Supply

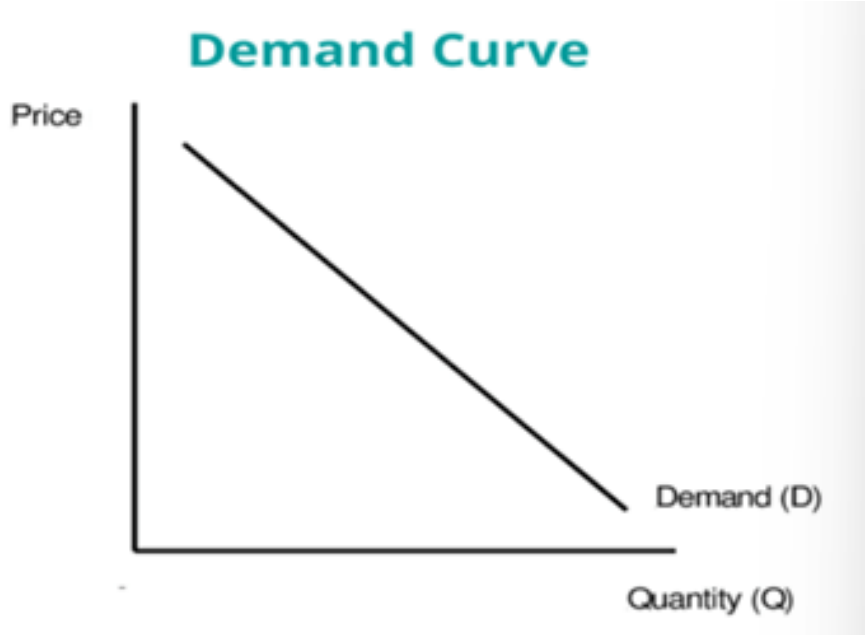
- Competitive Market
- Demand/Supply Curve
  - Shift vs. Movement Along Curve
- Equilibrium
  - Surplus & Shortage

Demand and Supply

- Competitive Market
  - Negligible impact on the market price → Price takers
  - All goods are exactly the same

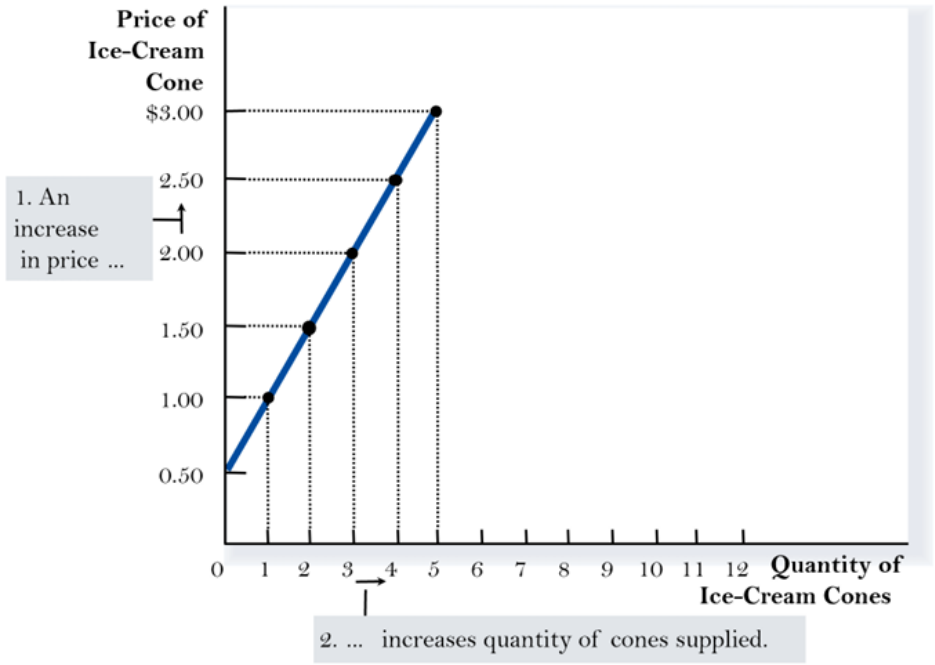
Demand and Supply

- Law of Demand
  - The quantity demanded of a good falls when the price of the good rises, other things equal



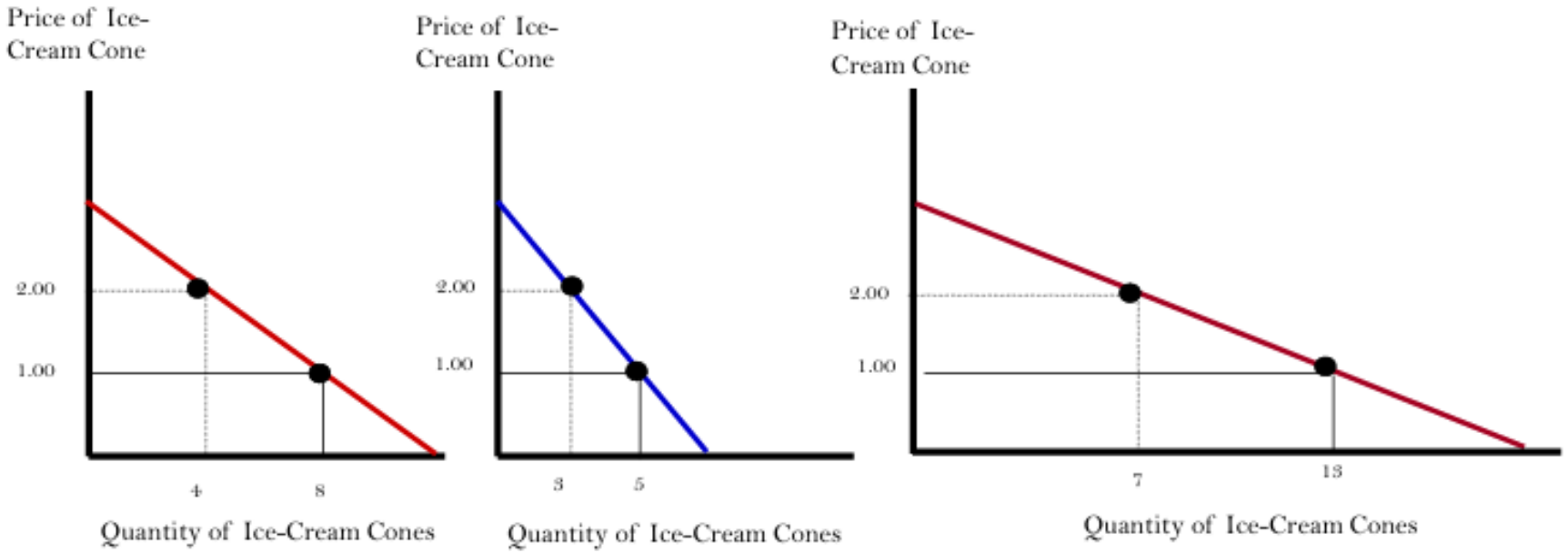
Demand and Supply

- Law of Supply
  - The quantity supplied of a good rises when the price of the good rises, other things equal



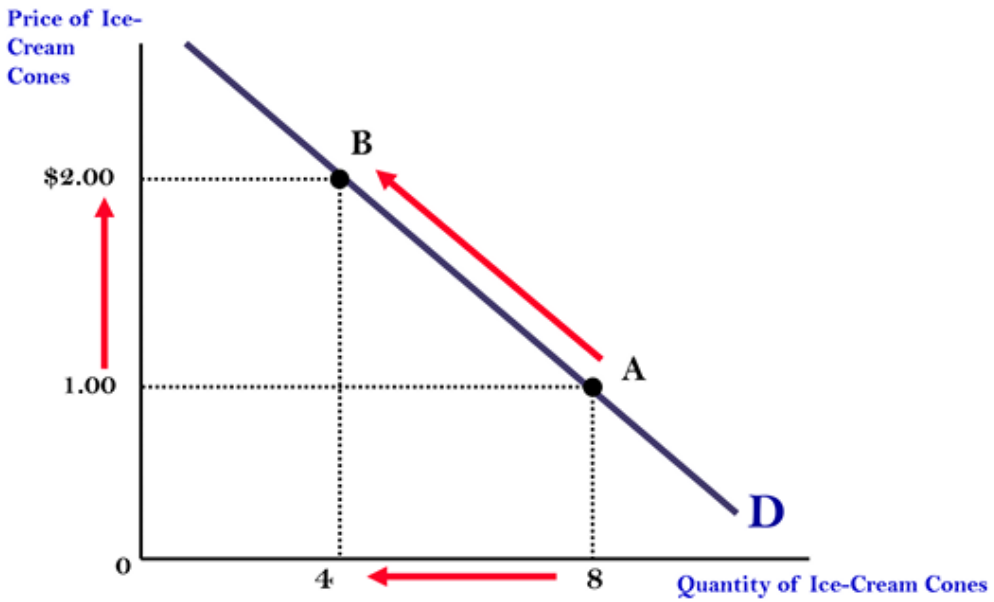
Demand and Supply

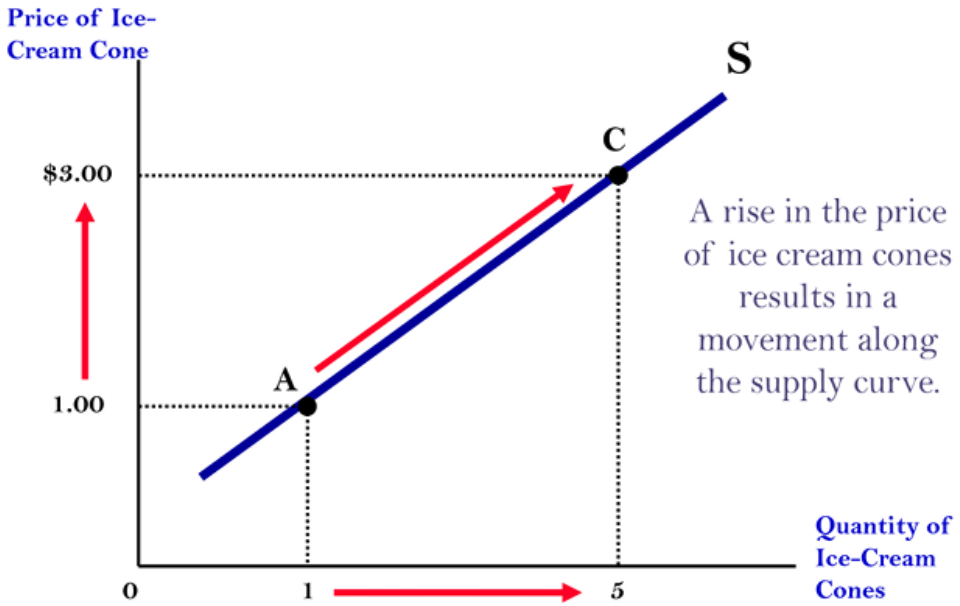
- Market Demand/Supply Curve:
  - Sum the individual demand curves **horizontally (price takers)**



Demand and Supply

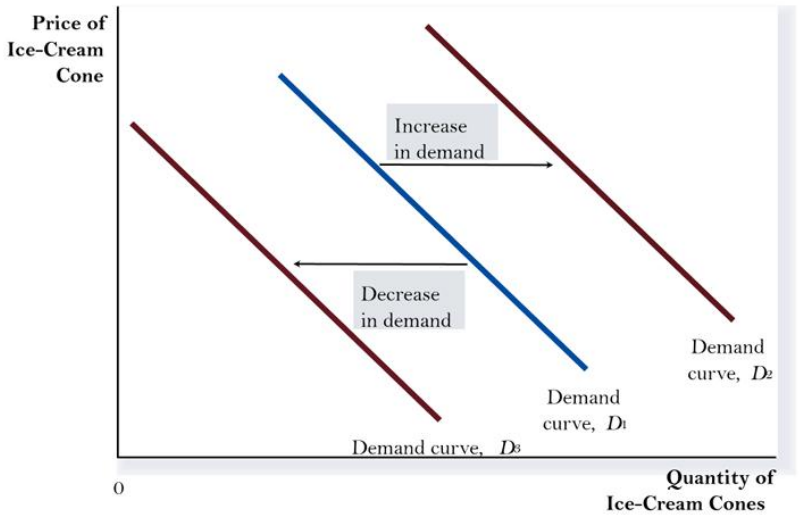
- Movement along the demand/supply curve
  - Caused by a **change in the price** of the product

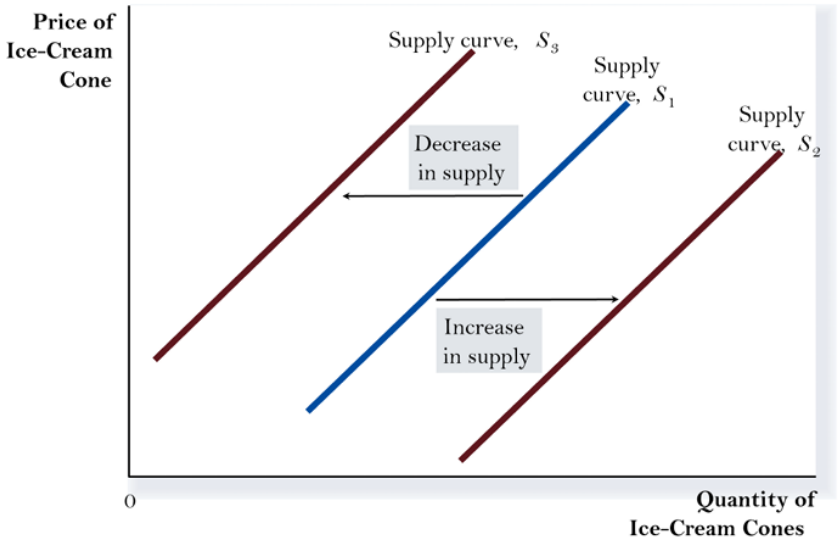




Demand and Supply

- Change in Demand/Supply (Shifts)
  - Caused by **non-price determinants** of demand/supply that determine demand/supply for a good





Demand and Supply

- Change in Demand (Shifts):
  - Pay attention to **substitutes and complements**

Variable	A Change in This Variable . . .
Price of the good itself	Represents a movement along the demand curve
Income	Shifts the demand curve
Prices of related goods	Shifts the demand curve
Tastes	Shifts the demand curve
Expectations	Shifts the demand curve
Number of buyers	Shifts the demand curve

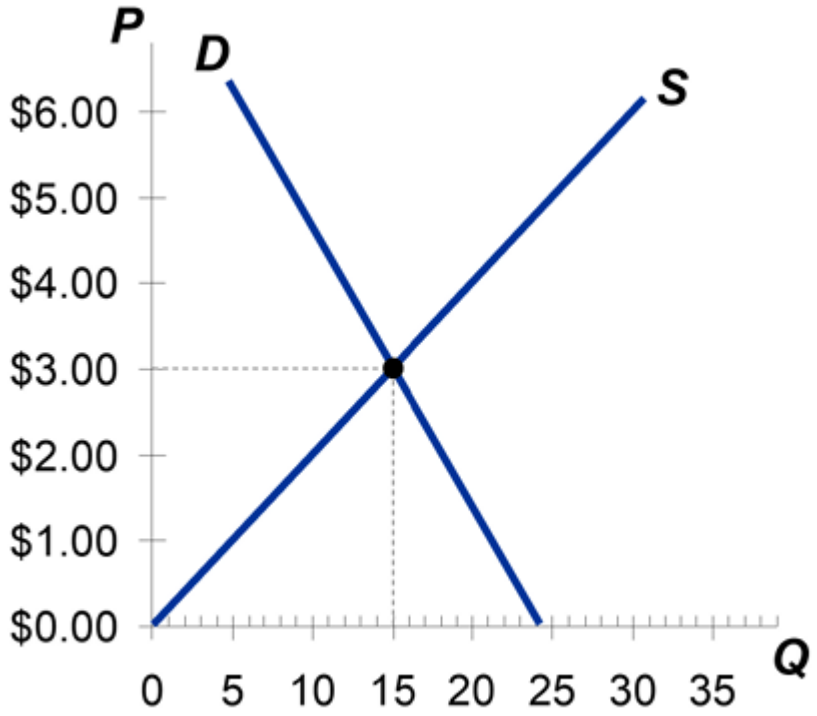
Demand and Supply

- Change in Supply (Shifts)

Variable	A Change in This Variable . . .
Price of the good itself	Represents a movement along the supply curve
Input prices	Shifts the supply curve
Technology	Shifts the supply curve
Expectations	Shifts the supply curve
Number of sellers	Shifts the supply curve

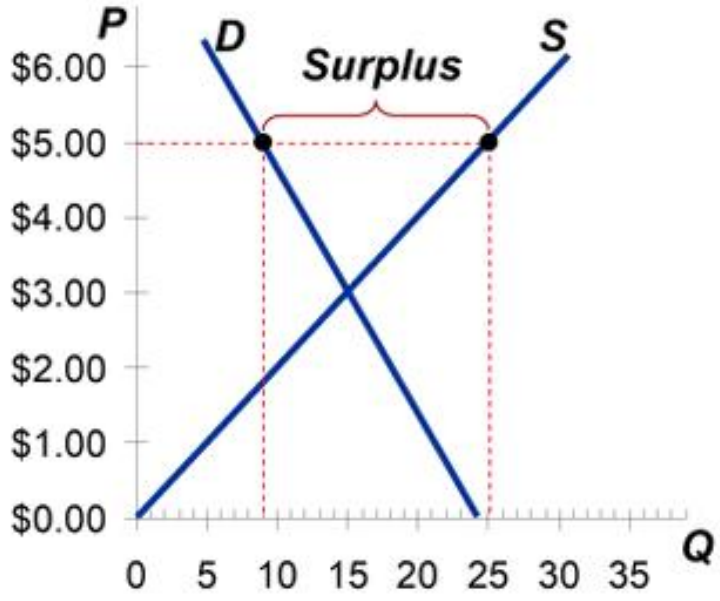
Demand and Supply

- Equilibrium: Price has reached the level where quantity supplied equals quantity demanded



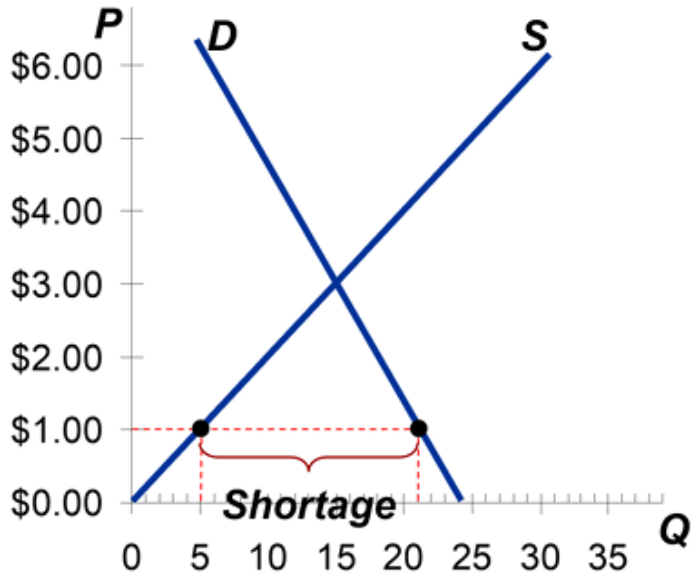
Demand and Supply

- Surplus (excess supply):
  - Quantity supplied is greater than quantity demanded



Demand and Supply

- Shortage (excess demand):
  - Quantity demanded is greater than quantity supplied



Demand and Supply

- Three steps to analyzing changes in equilibrium
  - Decide **whether the event shifts** the supply curve, the demand curve, or, in some cases, both curves
  - Decide whether the curve shifts to the **right or to the left**
  - Use the supply-and-demand diagram

    Compare the initial and the new equilibrium

    Effects on equilibrium price and quantity

Elasticity

- Price Elasticity of Demand/ Supply
- Midpoint method
- Determinants of elasticity
- Variety of elasticity
- Price elasticity and Total Revenue

Elasticity

- Elasticity
  - Measure of the responsiveness of Qd or Qs to a change in one of its determinants
    - (like price: price elasticity of demand): How much the quantity demanded of a good responds to a change in the price of that good
    - We will **drop the minus sign** and report all price elasticities as **positive** numbers

Elasticity

- Elasticity
  - Price elasticity of demand = $\dfrac{percentage \quad change \quad in \quad Q^d}{percentage \quad change \quad in \quad P}$
  - Calculation Methods
    - Standard method for (%) change $= \dfrac{end \quad value - start \quad value}{start \quad value} \times 100\%$
    - Midpoint Method:

$$\text{percentage change} = \frac{\text{end value - start value}}{\text{midpoint}} \times 100\%$$

$$\text{Price elasticity of demand} = \frac{(Q_2 - Q_1) / [(Q_2 + Q_1) / 2]}{(P_2 - P_1) / [(P_2 + P_1) / 2]}$$

Elasticity

- Demand Elasticity
  - is higher when **close substitutes are available**
  - is higher for **narrowly defined goods** than for broadly defined ones
  - is higher for **luxuries** than for necessities
  - is higher in the **long run**

Elasticity

- Supply Elasticity
  - is higher when **sellers can change the quantity they produce**
  - is higher in the **long run**

Elasticity

- Elasticity Variety (flatter → more elastic)
  - Elastic
  - Inelastic
  - Unit elasticity
  - Perfectly inelastic
  - Perfectly elastic
  - Note: The elasticity of the linear demand curve is not constant!

Elasticity

- Price Elasticity and Total Revenue
  - Total Revenue (TR) = P x Q
    - A price increase has two effects on revenue:
      - Higher revenue: because of the higher P
      - Lower revenue: you sell fewer units (lower Q)
    - If the demand is elastic, the % change in Q will be greater than the % change in price; otherwise, smaller

Consumer and Producer Surplus

- Willingness to Pay (WTP)
- Consumer Surplus
- Producer Surplus
- Total Surplus (and equilibrium)

Consumer and Producer Surplus

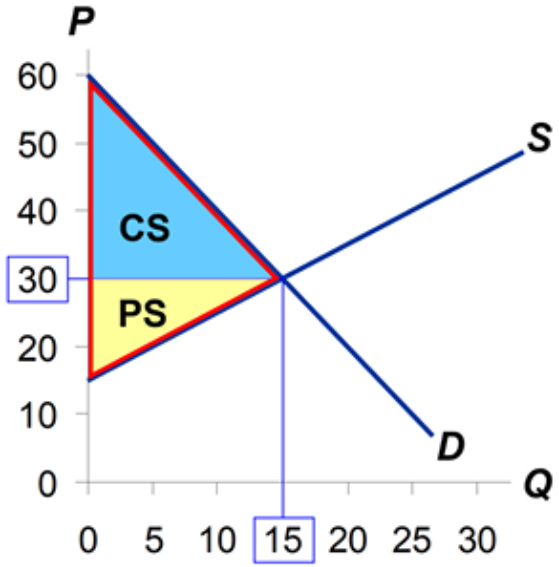
- Willingness to Pay (WTP)
  - The maximum amount the buyer will pay for that good
- Willingness to Sell (Cost)
  - Value of everything a seller must give up to produce a good

Consumer and Producer Surplus

- Consumer surplus CS = WTP- P
  - The area under the demand curve above the price, from 0 to Q
- Producer Surplus PS = P – cost
  - The area above the supply curve under the price, from 0 to Q
- Total surplus = CS + PS = Value to buyers – Cost to sellers
  - Measure of society's well-being

Consumer and Producer Surplus

- Consumer surplus CS = WTP- P
- Producer Surplus PS = P – cost
- Total surplus = CS + PS = Value to buyers – Cost to sellers



Government Intervention

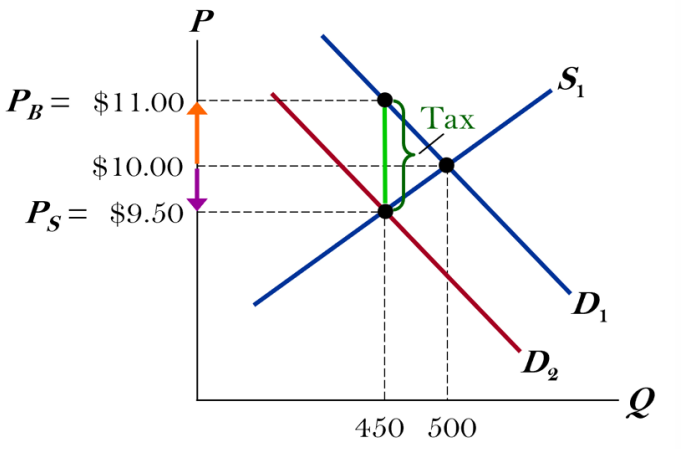
- Government Policies: ceiling and floor
- Taxes
  - Tax incidence
  - Elasticity and tax incidence
- Import quotas and tariffs

Government Intervention

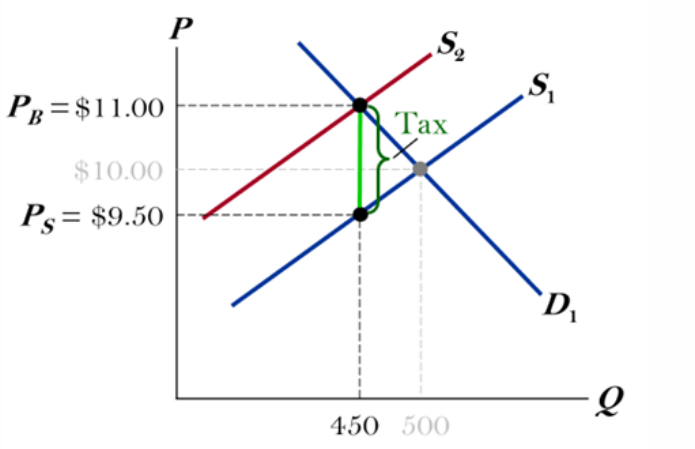
- Government Policies
  - Price controls (note: whether the ceiling or floor is binding)
    - Price Ceiling
    - Price Floor
  - Taxes
    - Tax incidence (how the burden of a tax is shared)

Government Intervention

- A Tax on Buyers
  - shifts the D curve down by the amount of the tax



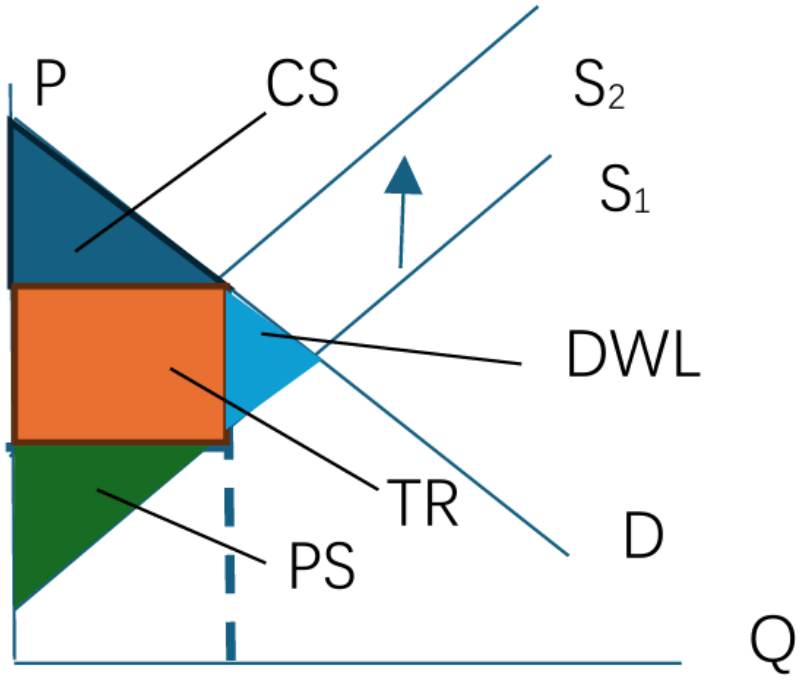
- A Tax on Sellers
  - shifts the S curve up by the amount of the tax



Tax incidence is the same!

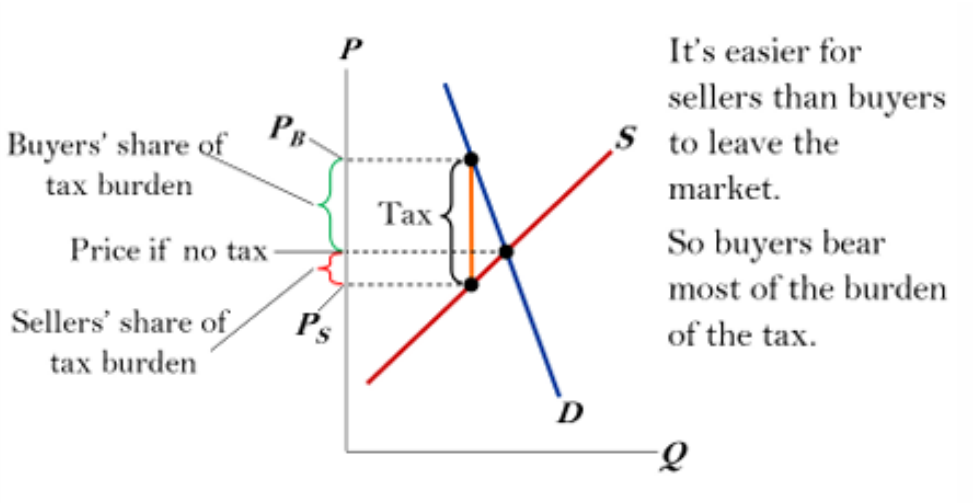
Government Intervention

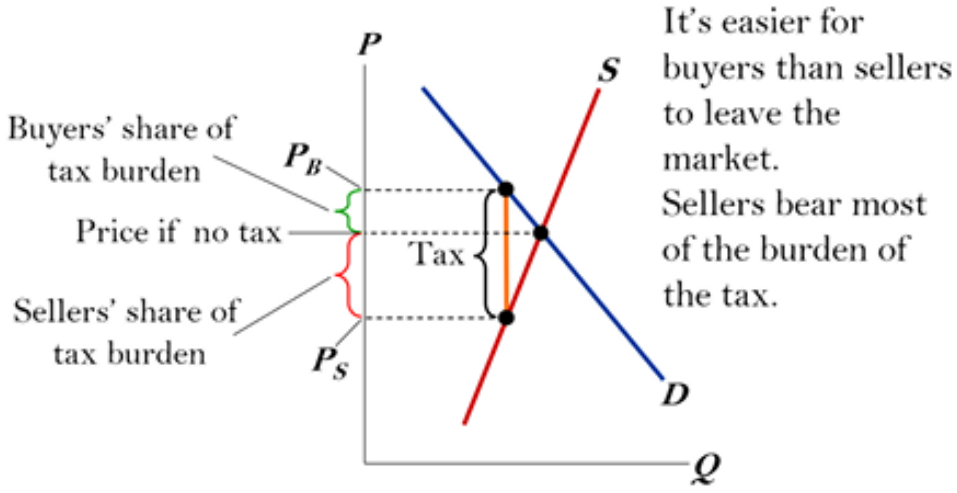
- A Tax on Sellers



Government Intervention

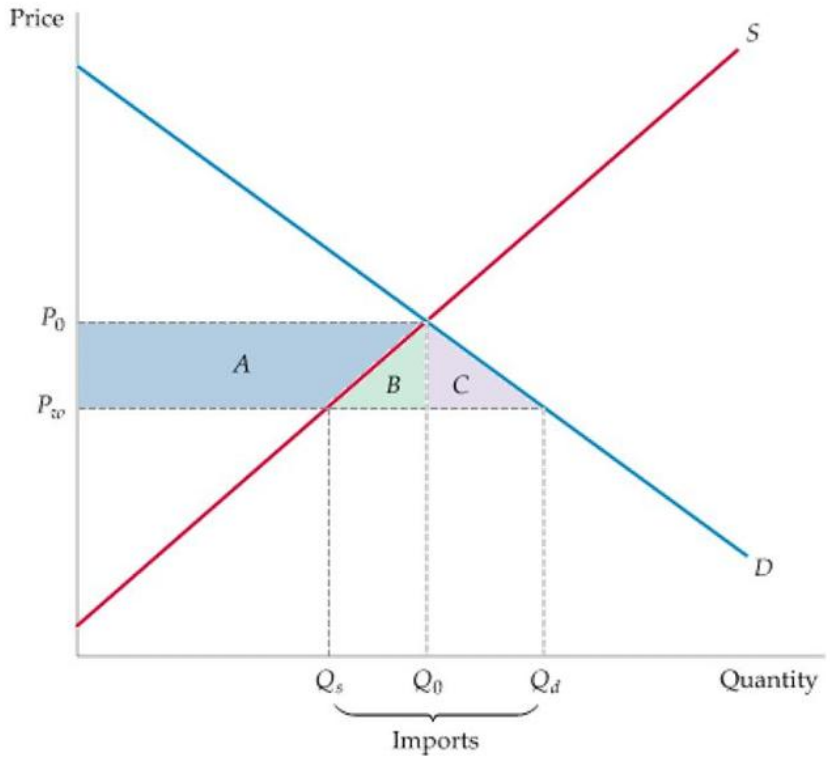
- Elasticity and Tax Incidence
  - **The party that is more elastic (flatter curve) bears less of the tax**
    - Because it's easier for the more elastic party to **leave the market**





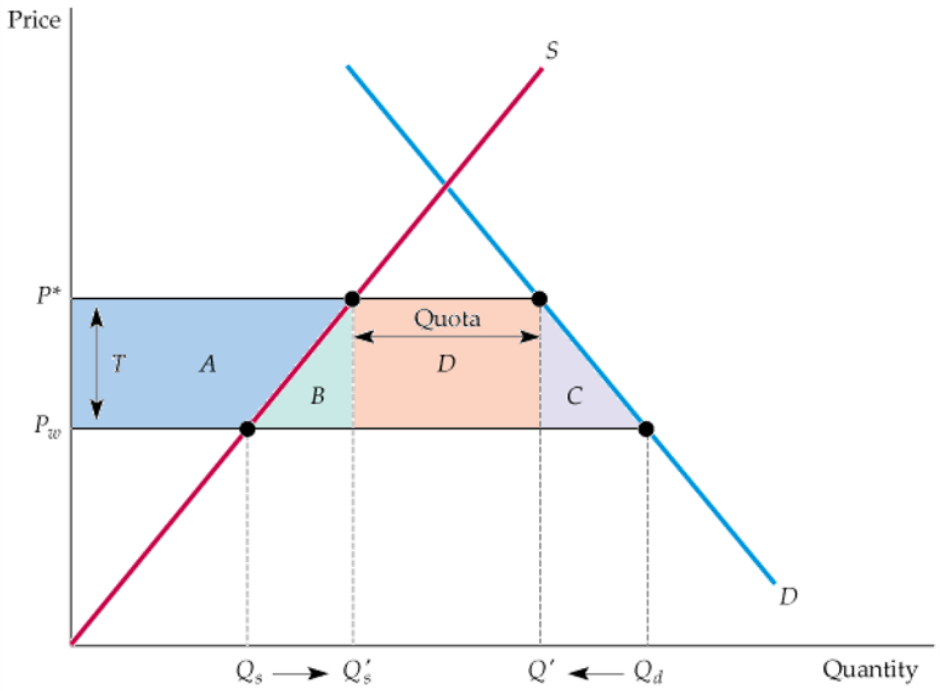
Government Intervention

- Import Tariff or Quota
  - Initially: A total Qd is consumed, Qs is supplied domestically, and the rest is imported
  - When imports are totally eliminated
    - The gain to producers: A
    - The loss to consumers: A + B + C
    - The deadweight loss is B + C



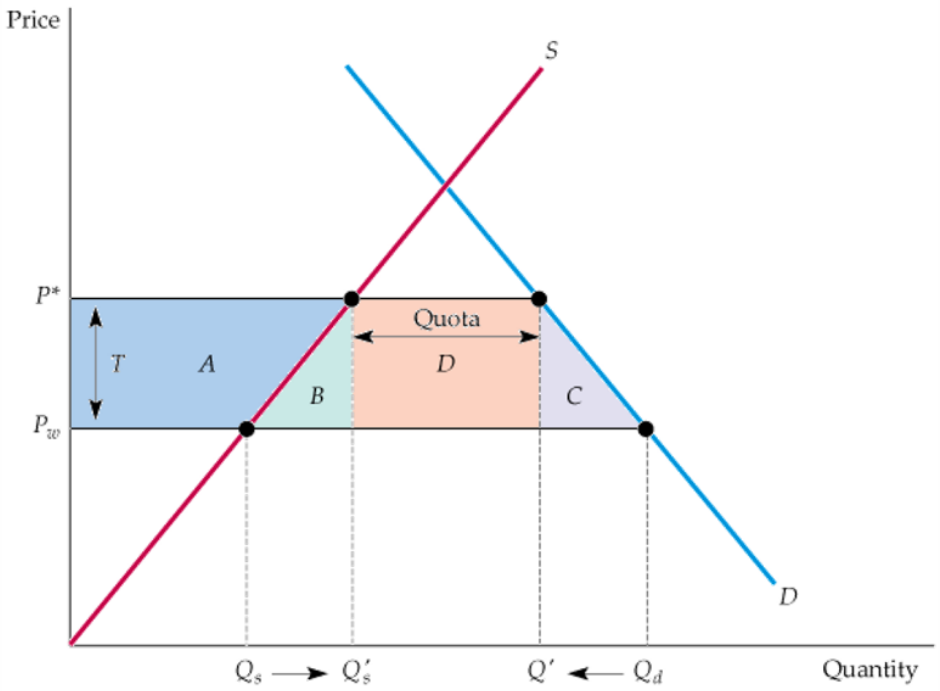
Government Intervention

- When imports are reduced
  - The gain to producers: A
  - The loss to consumers: A + B + C+D
    - Tariff:
      - Government gains: D
      - Net domestic loss: B + C



Government Intervention

- When imports are reduced
  - The gain to producers: A
  - The loss to consumers: A + B + C+D
    - Quota
      - Profits of foreign producers: D
      - Net domestic loss: B+C+D



→ The import tariff will result in **higher TS** than the import quota

Externality

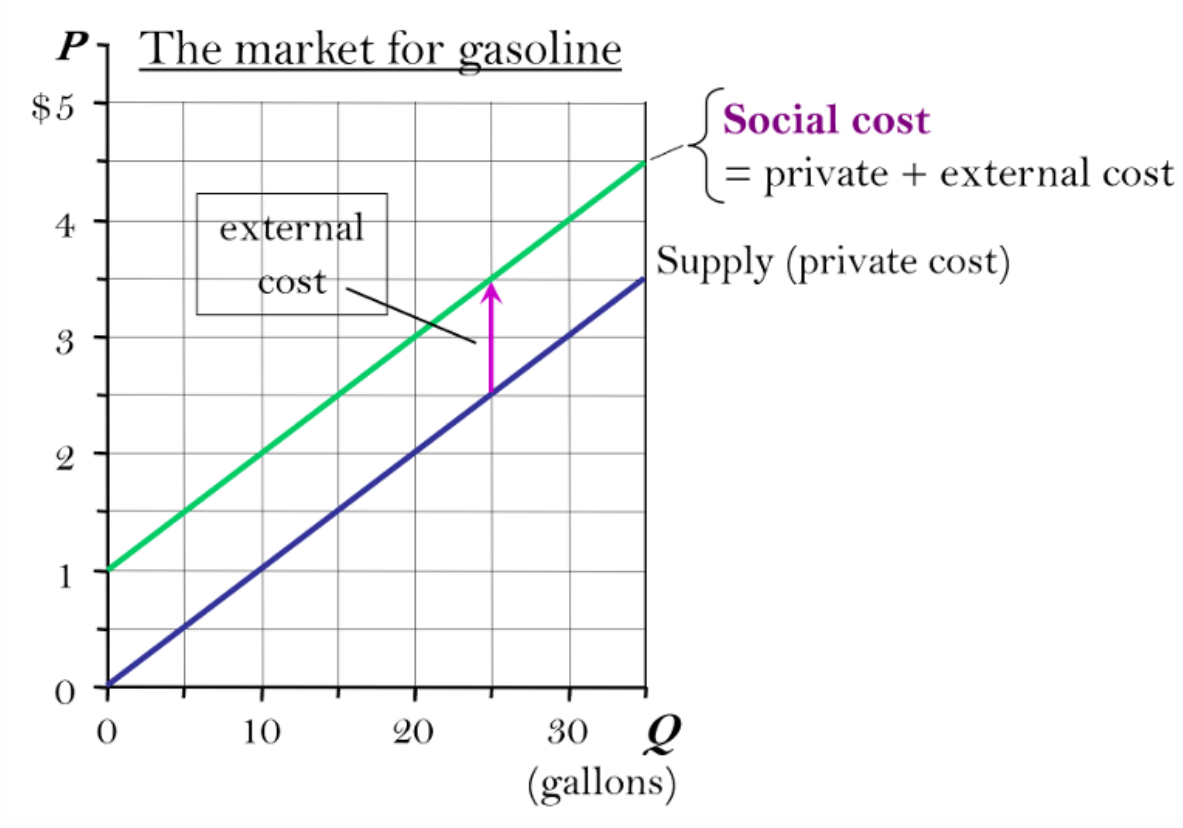
- Definition, negative and positive, examples
- Public policies toward externalities
  - Corrective taxes and subsidies
  - Internalizing the externality
- Coase theorem (private solutions to externalities)
  - Whatever the initial distribution of rights

Externality

- Externality: one type of market failure
  - The uncompensated impact of one person's actions on the well-being of a bystander
    - Negative externality
    - Positive externality

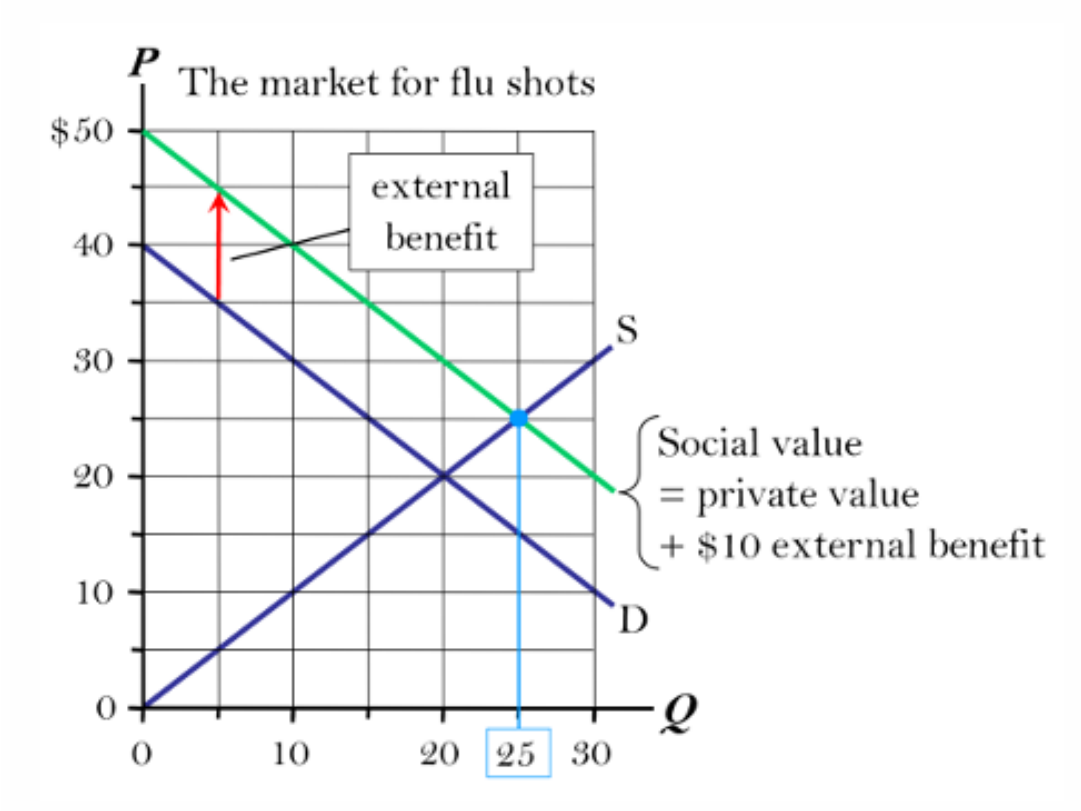
Externality

- Externality Cost = value of the negative impact on bystanders



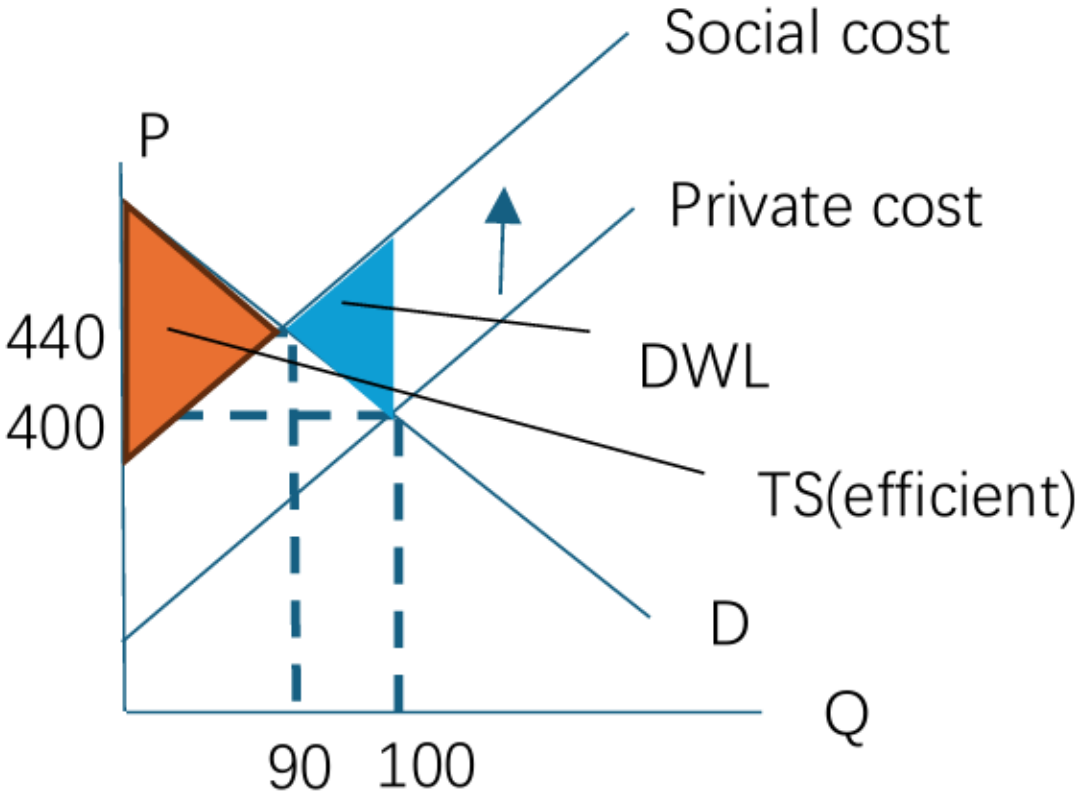
Externality

- Externality Benefit = value of the positive impact on bystanders



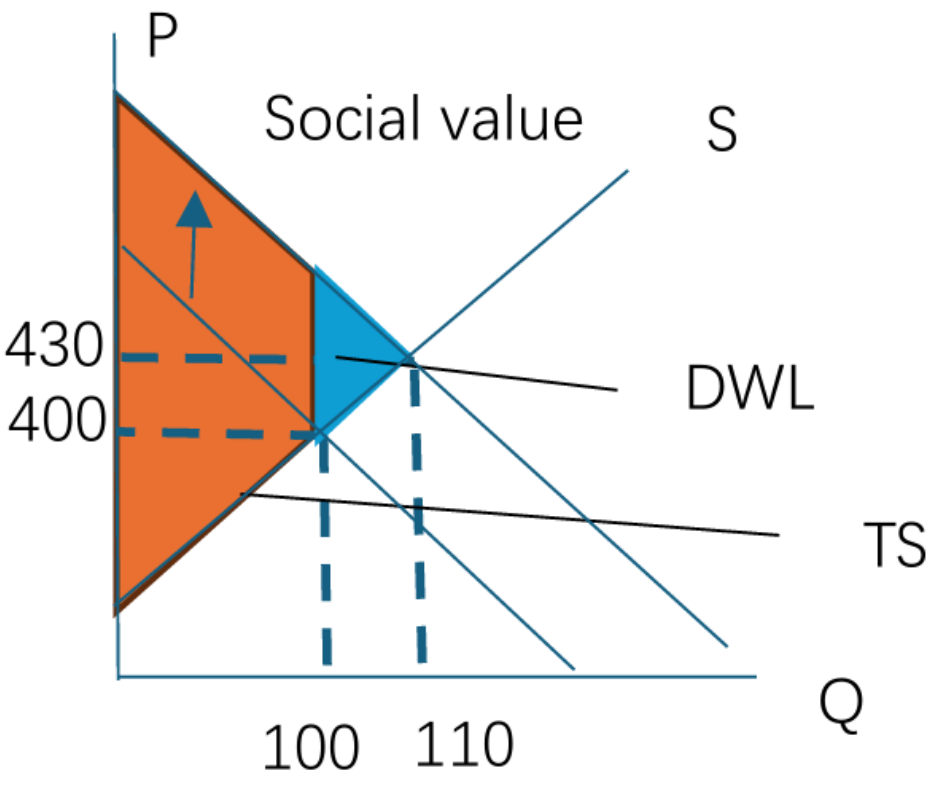
Externality

- Negative externalities
  - TS = TS (efficient)-DWL
  - DWL:
    - This triangle represents the social loss due to the **excess production** (Qe→Qc)



Externality

- Positive externalities
  - TS = TS (efficient)-DWL
  - DWL:
    - This triangle represents the social loss due to the **lower production** (Qe→Qc)



Externality

- Solutions
  - Public Policies
    - Command-and-control policies (regulate behaviors directly)
    - Market-based policies (Internalize the externality)
      - Corrective Taxes → Negative Externalities
      - Corrective Subsidies → Positive Externalities

Externality

- Solutions
  - Private Solutions: The Coase theorem
    - Whatever the initial distribution of rights
      - Everyone is better off with the overall efficient outcome
    - Do not work
      - Transaction costs
      - Coordination problems

Consumer Behaviors

- Budget Constraint
- Preferences
- Indifference curves—Utility and utility function
- Marginal Rate of Substitution (MRS)
- Perfect/Close Substitutes/Complements
- Optimization, Corner solutions
- Marginal Utility and Consumer Choice

Consumer Behaviors

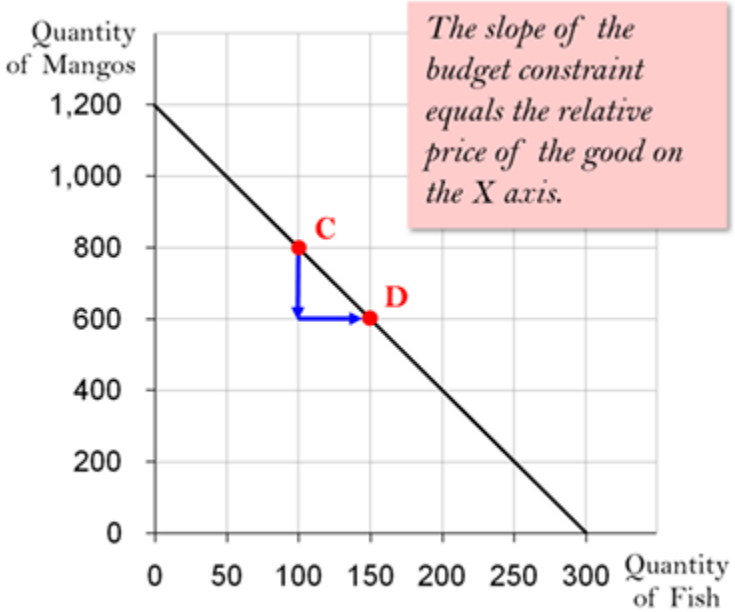
- Budget Constraint
  - The limit on the consumption bundles that a consumer can afford
- Budget Line $$P_A A + P_B B = I$$

Consumer Behaviors

- The Slope of the Budget Constraint
  - Relative price of the good on the X axis in terms of the Y axis good

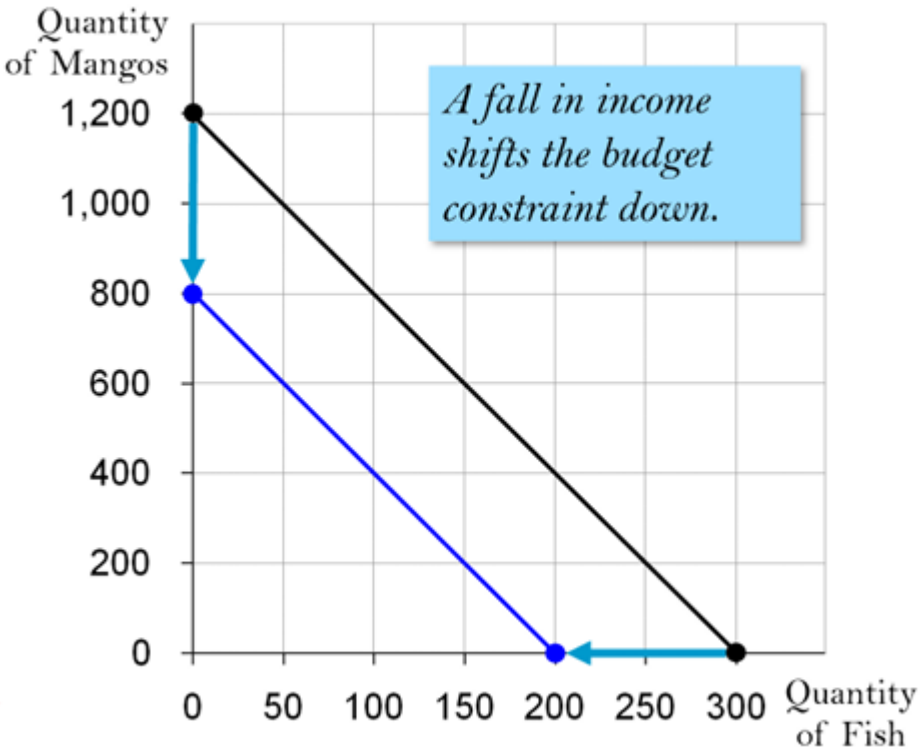
Must give up how many y-axis goods to get one x-axis good

Slope = Px/Py



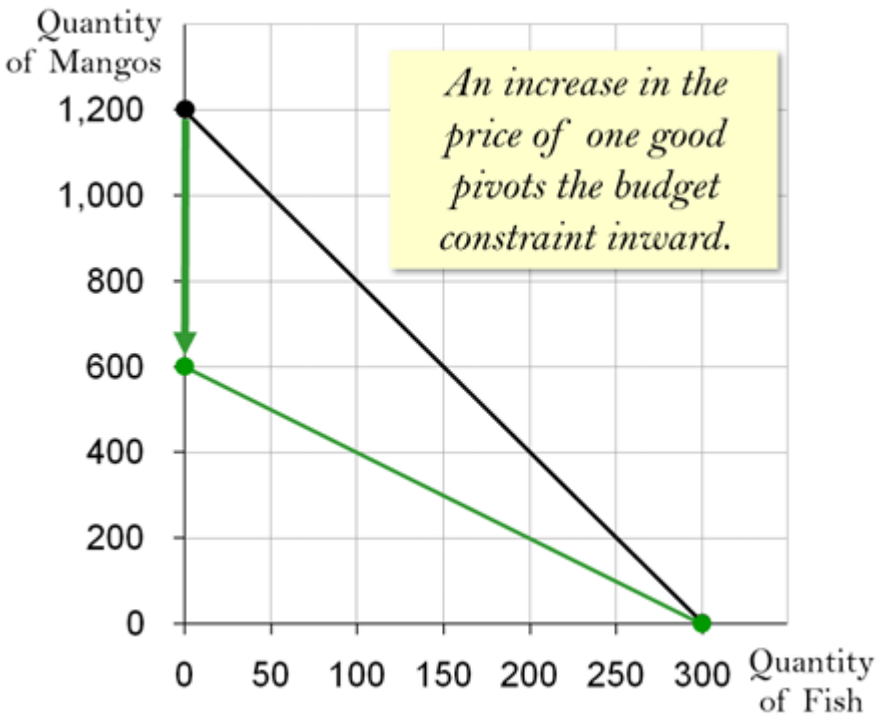
Consumer Behaviors

- The Shift of the Budget Constraint. A change in the price of one good
  - Change in income



Consumer Behaviors

- The Pivots of the Budget Constraint
  - A change in the price of one good

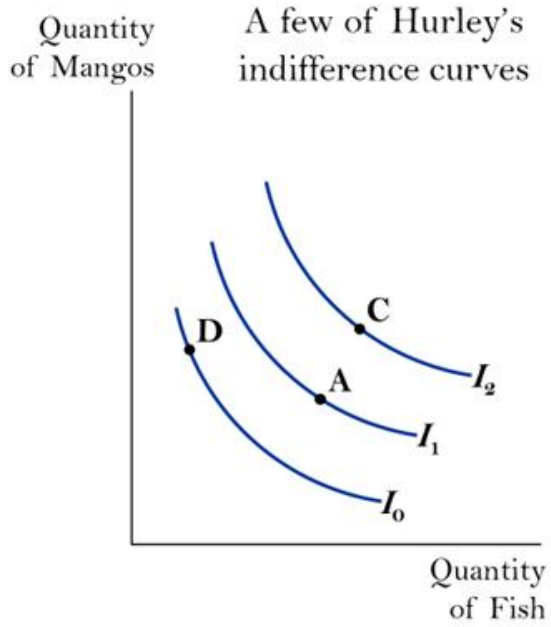


Consumer Behaviors

- Preferences: Basic assumptions
  - Completeness
    - Consumers can compare and rank all possible baskets
  - Transitivity
    - Preferences are transitive
    - If A > B, B > C, then A > C
  - More is better than less

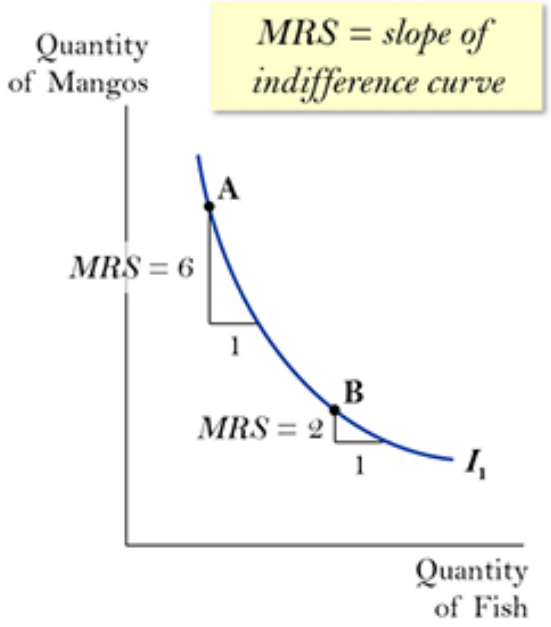
Consumer Behaviors

- Indifference Curve: Properties:
  - Indifference curves are downward sloping
  - Higher indifference curves are preferred to lower ones
  - Indifference curves cannot cross
  - Indifference curves are bowed inward



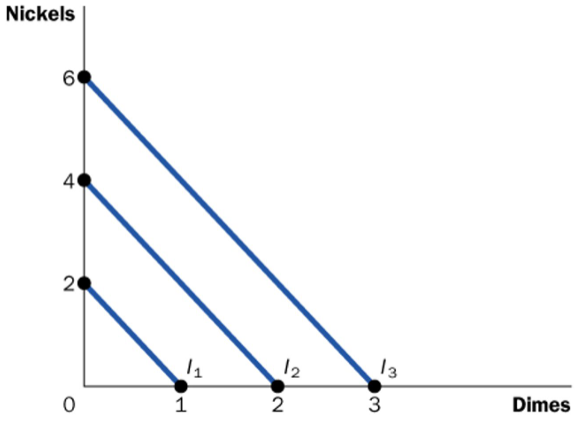
Consumer Behaviors

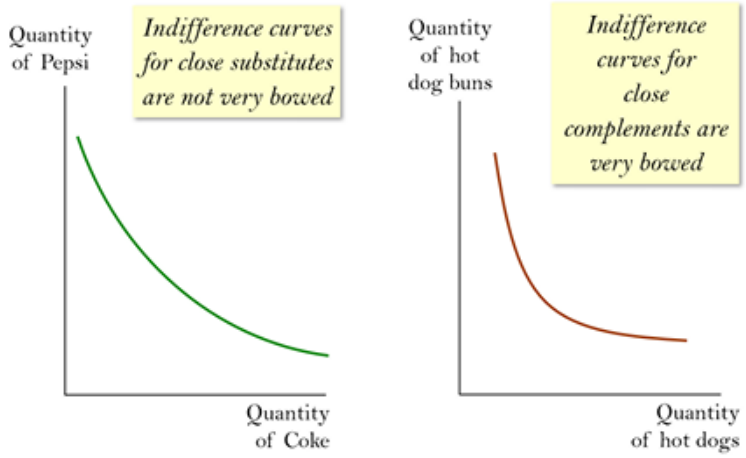
- Indifference Curve: Slope - Marginal rate of substitution (MRS):
  - The maximum amount of a good that a consumer is willing to give up to obtain one additional unit of another good
    - $MRS = \frac{MU_x}{MU_y} = \frac{du/dx}{du/dy}$
  - Convexity
    - Diminishing MRS

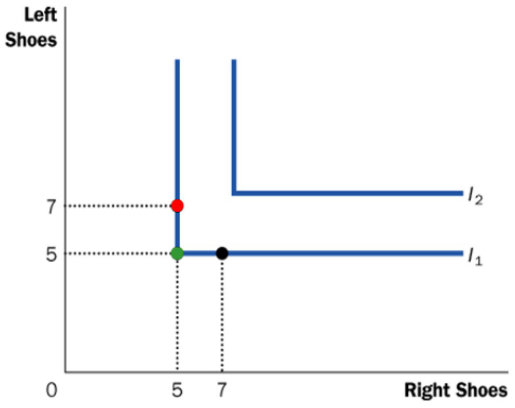


Consumer Behaviors

- Indifference Curve: Extreme Cases
  - Perfect Substitutes
  - Perfect Complements
  - Close Substitutes and Close Complements

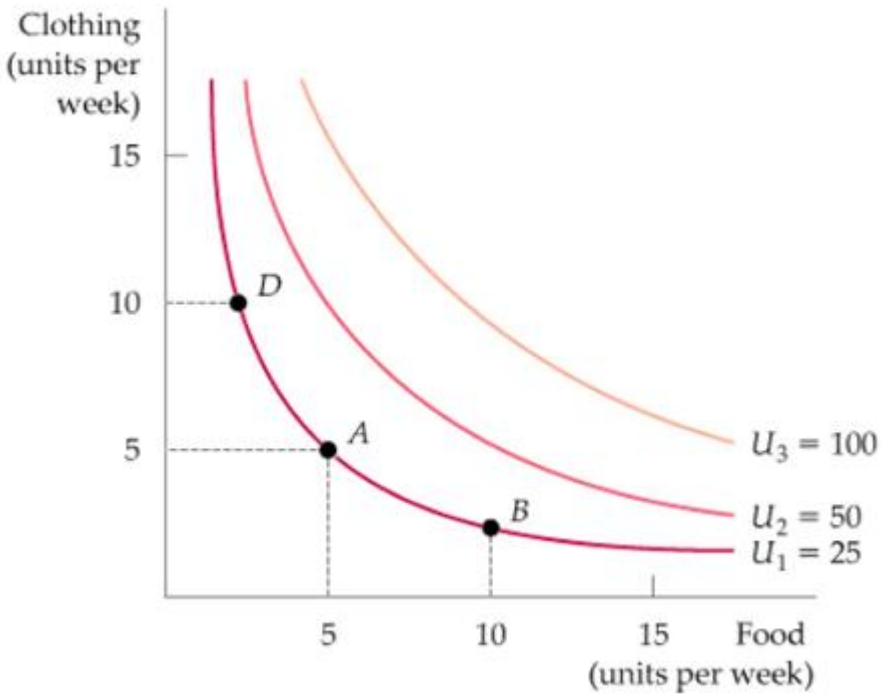






Consumer Behaviors

- Utility and Utility Function
  - Numerical score representing the satisfaction that a consumer gets from a given bundle



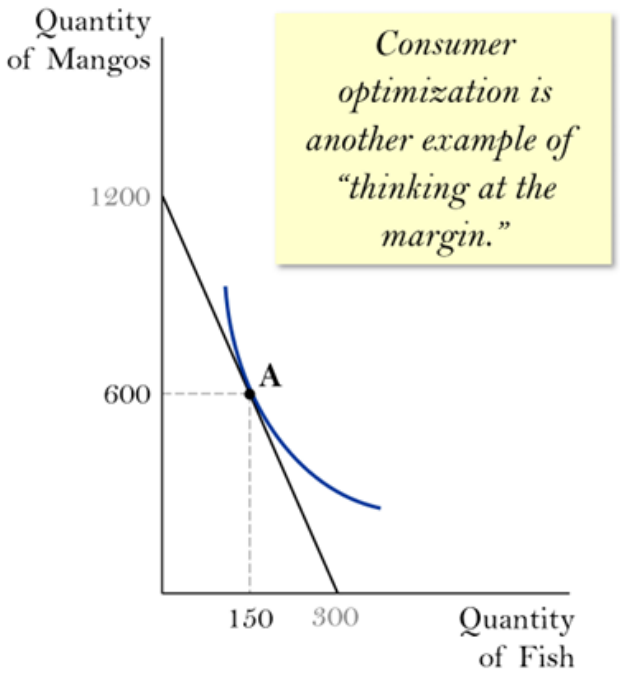
Consumer Behaviors

- Optimization (given budget constraint and utility function)
  - Optimum: the point on the budget constraint that touches the highest possible indifference curve.

$$\text{MRS} = P_F / P_M$$

marginal value of fish (in terms of mangos)

price of fish (in terms of mangos)



Consumer Behaviors

- Marginal Utility and Consumer Choice

$$\text{MRS} = P_F / P_C$$

$$\text{MU}_F / \text{MU}_C = P_F / P_C$$

$$\text{MU}_F / P_F = \text{MU}_C / P_C$$

- Equal marginal principle:
  - Utility is maximized when the consumer has equalized the marginal utility per dollar of expenditure across all goods

Consumer Behaviors

- Question
  - Utility function: u (x, y) = f (x, y), Px, Py, I
    - MRS:

$$\mathrm{MRS}_{xy} = \frac{\partial u/\partial x}{\partial u/\partial y} = \frac{\frac{\partial f(x,y)}{\partial x}}{\frac{\partial f(x,y)}{\partial y}} = \frac{MU_x}{MU_y}$$

    - Budget Constraint

$$P_x \cdot x + P_y \cdot y = I$$

    - Optimal Consumption

(x > 0, y > 0)

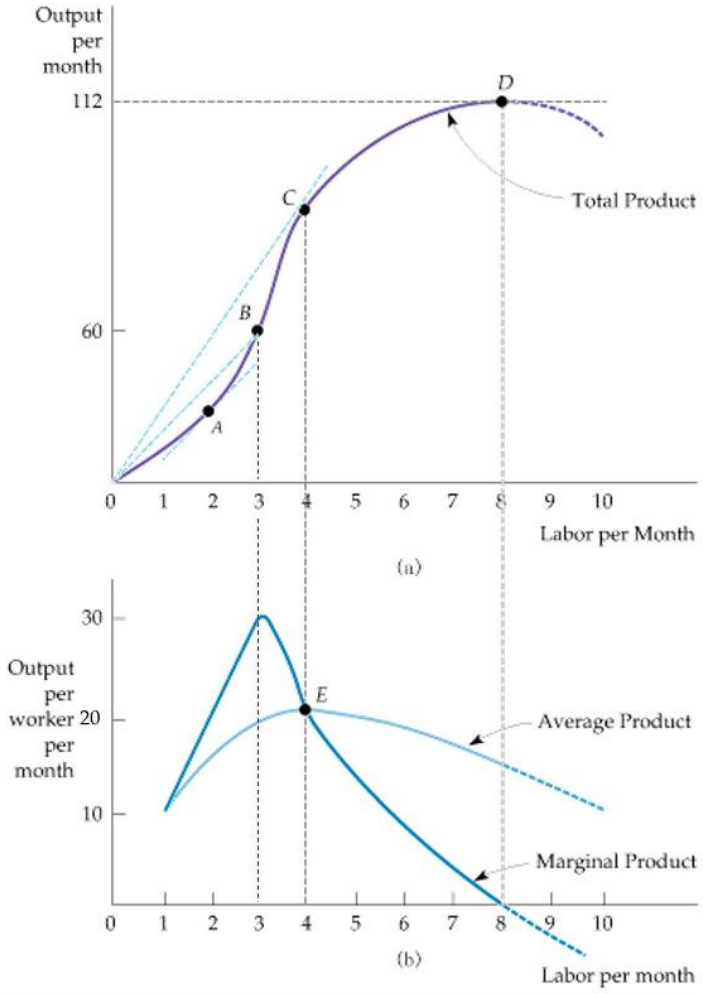
$$\begin{cases} \frac{\partial f(x,y)/\partial x}{\partial f(x,y)/\partial y} = \frac{P_x}{P_y} \\ P_x \cdot x + P_y \cdot y = I \end{cases}$$

Production and Cost

- Production Function
- Marginal Rate of Technical Substitution (MRTS)
- Cost: Explicit vs. Implicit
- Economic Profit vs. Accounting Profit
- MC, FC, VC, AFC, AVC, ATC
- Costs in the Short Run & Long Run
- Choosing Inputs

Production and Cost

- Production function
  - q = F (K, L)
  - Single input: Marginal product of labor
    - MPL = ΔQ/ΔL, **Diminishing**
  - Average Product
    - APL = q/L
  - MP > AP, AP rises; MP < AP, AP falls
    - They intersect at AP's peak

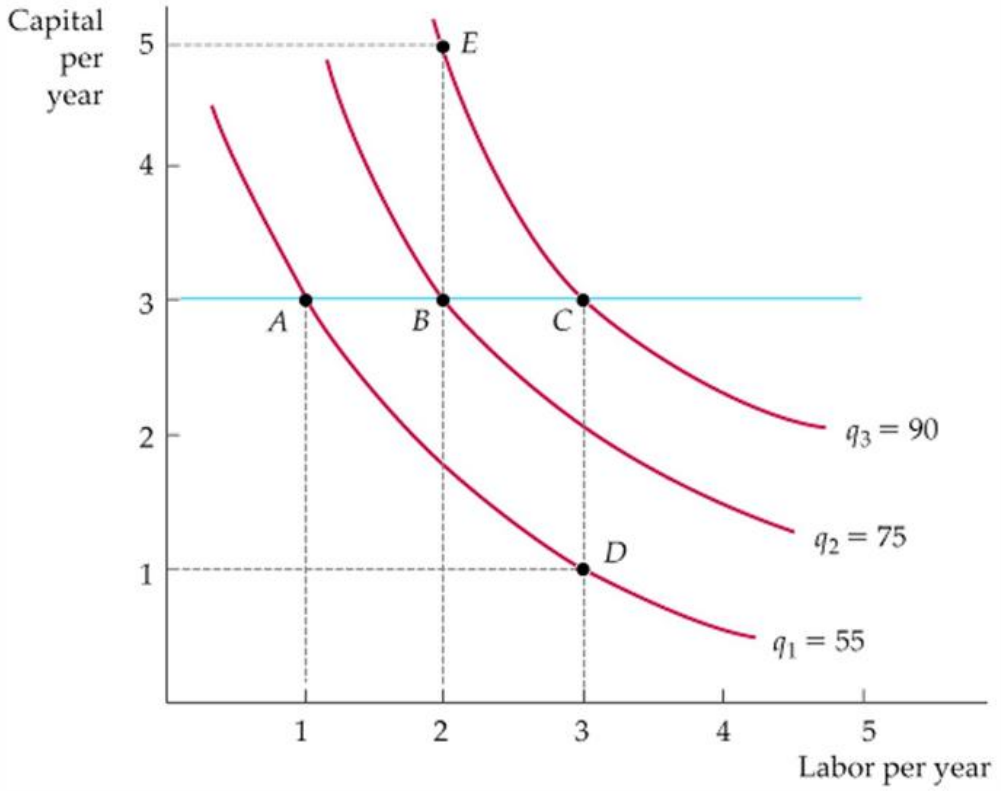


Production and Cost

- Production with Two Variable Inputs (Isoquants)
  - Marginal Rate of Technical Substitution (MRTS) , **Diminishing**

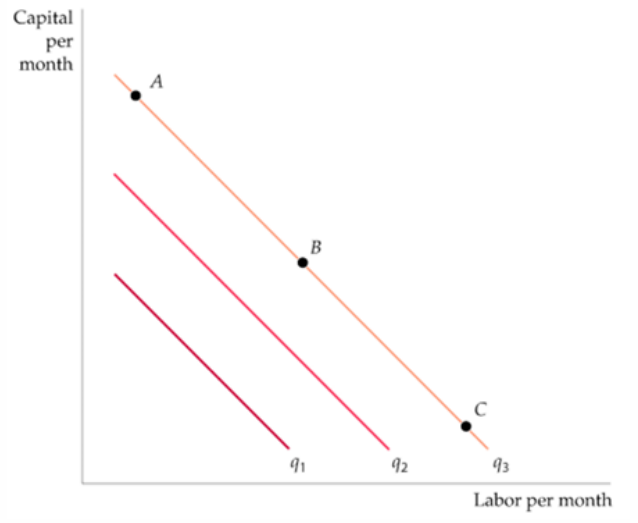
= − Change in capital input/change in labor input

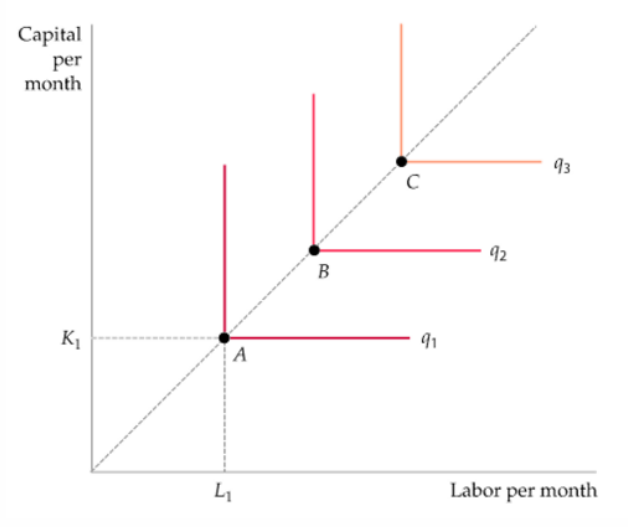
$$\text{MRTS} = -\Delta K/\Delta L = \text{MP}_L/\text{MP}_K$$



Production and Cost

- Production with Two Variable Inputs: Two special cases
  - Perfect Substitutes
  - Fix-proportions





Production and Cost

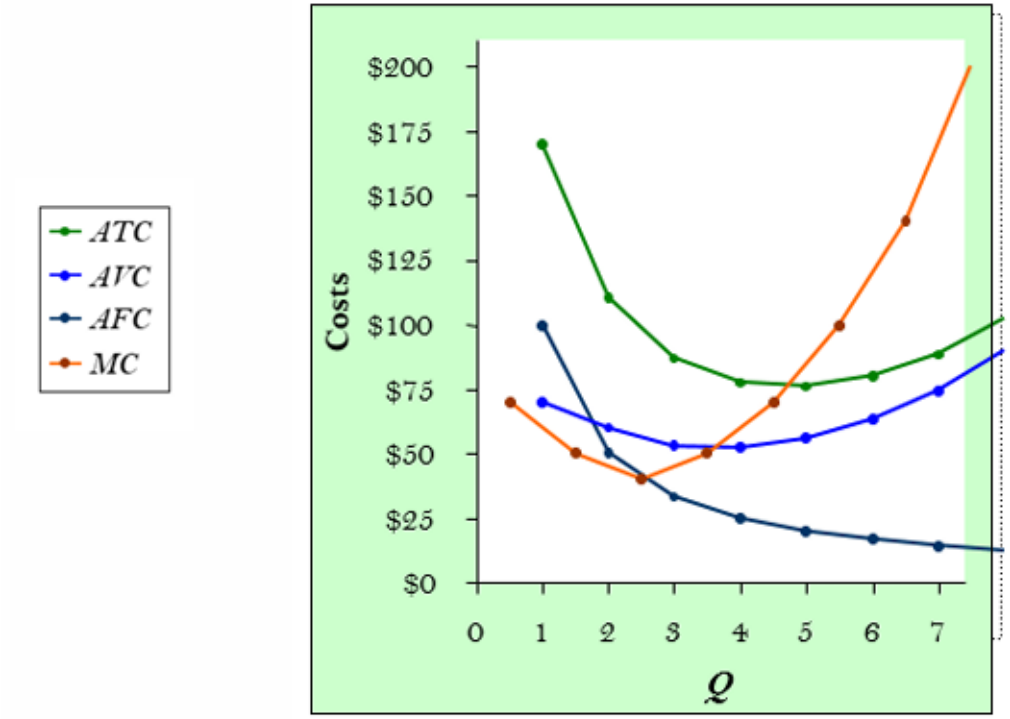
- Explicit Costs
  - Require an outlay of money
- Implicit Costs
  - Do not require a cash outlay
- Total cost = Explicit + Implicit costs
- Accounting Profit
  - TR-Total explicit costs
- Economic Profit
  - TR-Total costs (explicit and implicit)

Production and Cost

- Marginal cost, **MC** = $\Delta TC/\Delta Q$
  - Increase in total cost arising from an extra unit of production
- Fixed costs, **FC**
  - do not vary with the quantity of output produced
- Variable costs, **VC**
  - vary with the quantity of output produced
- Total cost = Fixed cost + Variable cost → **ATC = AFC + AVC**

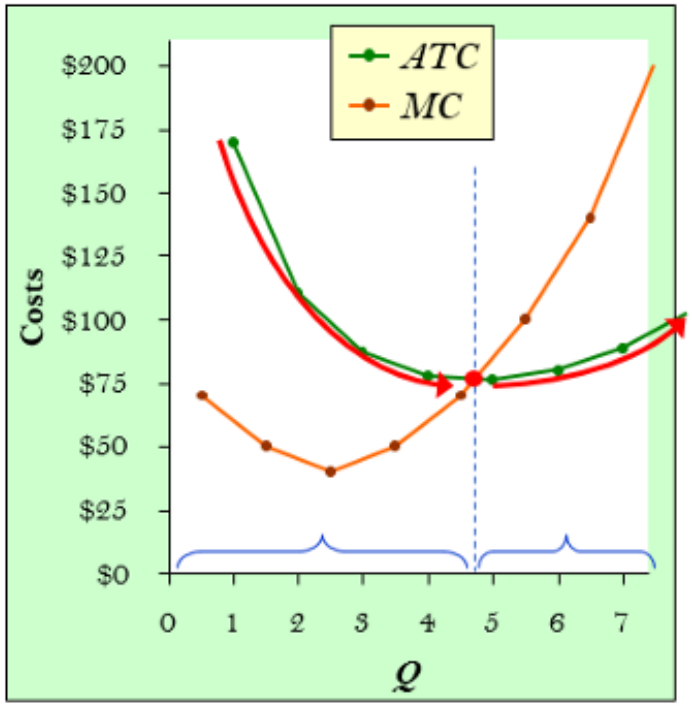
Production and Cost

- The Various Cost Curves Together
  - ATC is U-shaped due to falling AFC and rising AVC



Production and Cost

- ATC and MC
  - The MC curve crosses the ATC curve at the ATC curve's minimum



Production and Cost

- Costs in the Short Run & Long Run
  - Short run: Some inputs are fixed → FC
    - $\Delta VC = w\Delta L \rightarrow MC = \Delta VC/\Delta q = w\Delta L/\Delta q$

$$MC = w/MP_L$$

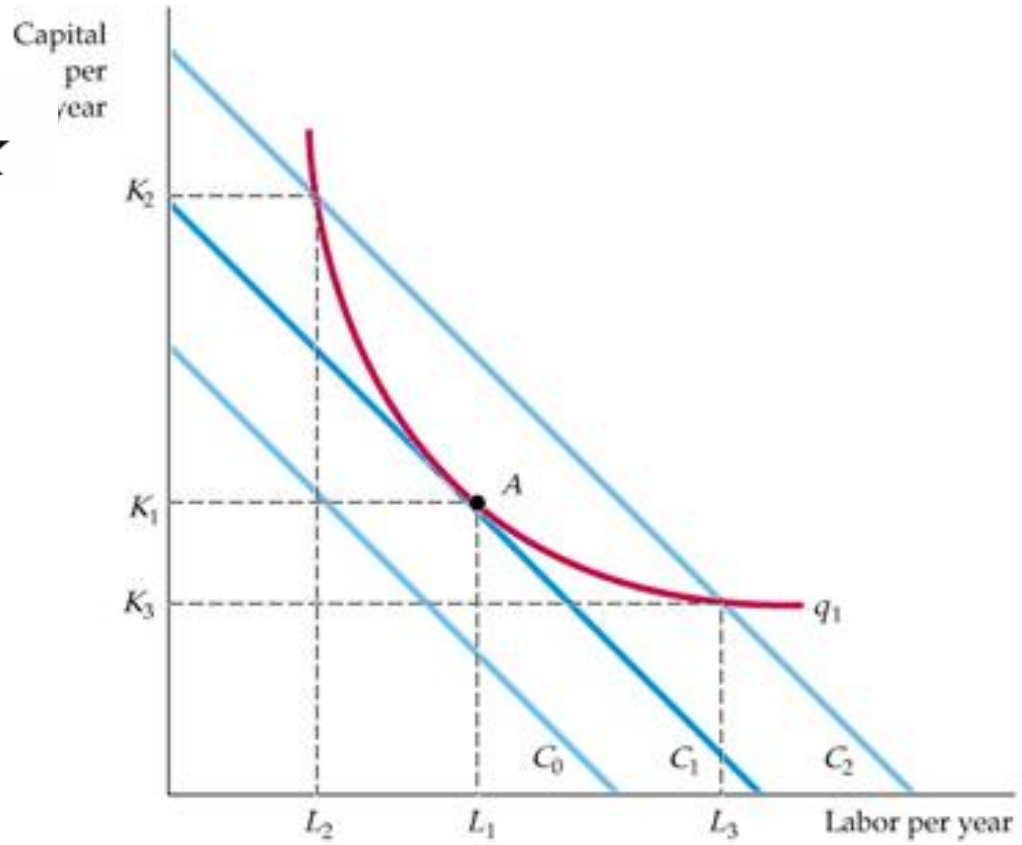
  - Long run: All inputs are variable
    - $C = wL + rK$ (Isocost)
    - **Slope = Px/Py =** w/r

Production and Cost

- Choosing Inputs in LR
  - $\text{MRTS} = -\Delta K/\Delta L = \text{MP}_L/\text{MP}_K$
  - When a firm minimizes the cost of producing a particular output

$$\text{MP}_L/\text{MP}_K = w/r$$

$$\text{MP}_L/w = \text{MP}_K/r$$



Production and Cost

- Question

Consider the (short-run) production function $q = F(L, \bar{K}) = \sqrt{L \cdot \bar{K}}$, where $\bar{K} = 1$. The total cost $C = wL + r\bar{K}$, where $w = r = 10$.

SR cost function:

$$L = q^2;\ C = 10L + 10 = 10(q^2 + 1)$$

$$MC = \frac{\Delta C}{\Delta q} = 20q;\ FC = r\bar{K} = 10; AFC = \frac{10}{q}; VC = wL = 10q^2;$$

$$AVC = \frac{10q^2}{q} = 10q;\ ATC = \frac{C}{q} = 10q + \frac{10}{q}$$

Production and Cost

- Question

Consider the (short-run) production function $q = F(L, \bar{K}) = \sqrt{L \cdot \bar{K}}$, where $\bar{K} = 1$. The total cost $C = wL + r\bar{K}$, where $w = r = 10$.

LR cost function: (K is variable rather than fixed)

$$\text{MRTS}=\frac{MP_L}{MP_K} = \frac{\Delta q/\Delta L}{\Delta q/\Delta K} = \frac{\frac{1}{2}L^{-\frac{1}{2}}K^{\frac{1}{2}}}{\frac{1}{2}L^{\frac{1}{2}}K^{-\frac{1}{2}}} = \frac{K}{L}$$

$$\frac{K}{L} = \frac{w}{r} = 1 \rightarrow \mathrm{K} = \mathrm{L} = q \rightarrow C = 20q$$

Competitive Markets

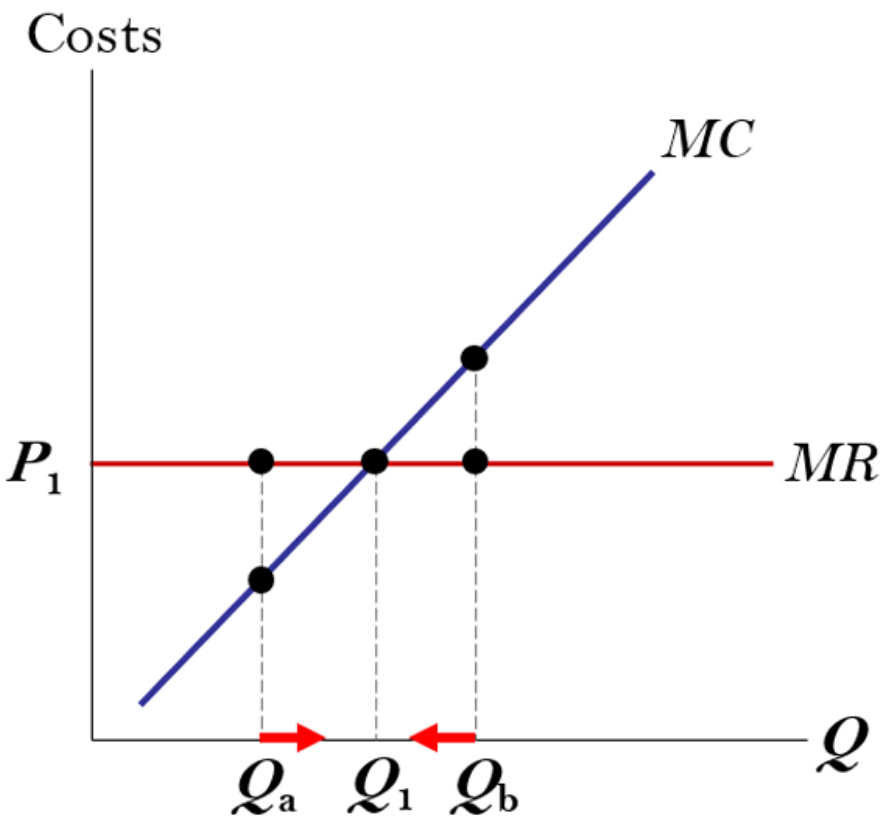
- Competitive Market
- MR = P
- Profit Maximization
- MC and the Firm's Supply Decision
- Shutdown vs. Exit
- SR/LR Supply Curve
- The Zero-Profit Condition

Competitive Markets

- $TR = P*Q$
- $AR = TR/Q = P*Q/Q = P$
- $MR = \Delta TR / \Delta Q$
- **AR = P = MR (price taker)**

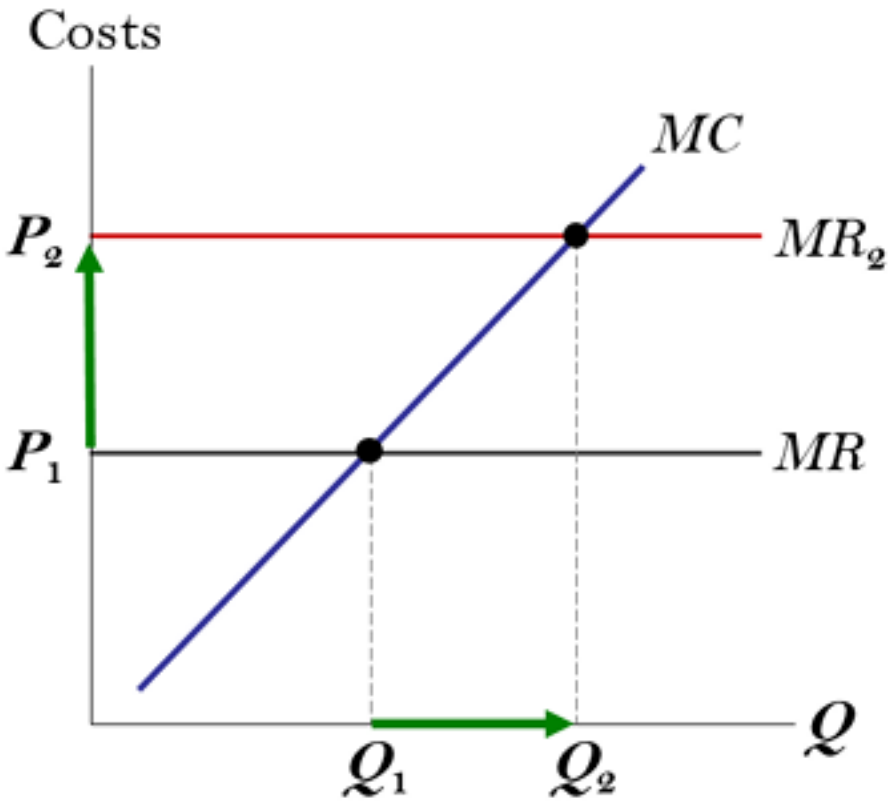
Competitive Markets

- Profit Maximization
  - **MR = MC**



Competitive Markets

- Profit Maximization
  - **MR = MC**
  - The MC curve determines the firm's Q at any price → Can change the y axis to price



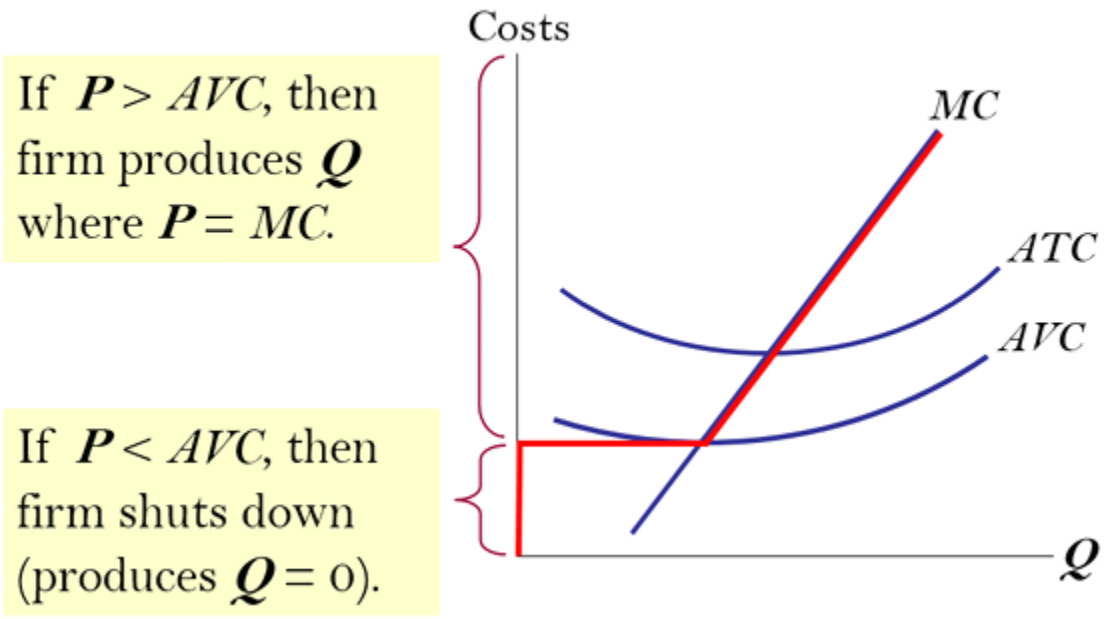
MC curve is the firm's supply curve!

Competitive Markets

- **Shut down (TR < VC, or P < AVC)**
  - A **short-run decision** not to produce because of market conditions
  - Zero VC, but still pay FC
- **Exit (TR < TC, or P < ATC)**
  - A **long-run decision** to leave the market
  - Zero costs
- The number of firms in the market is **fixed in the short run** (due to fixed costs), but variable in the long run (due to free entry and exit)

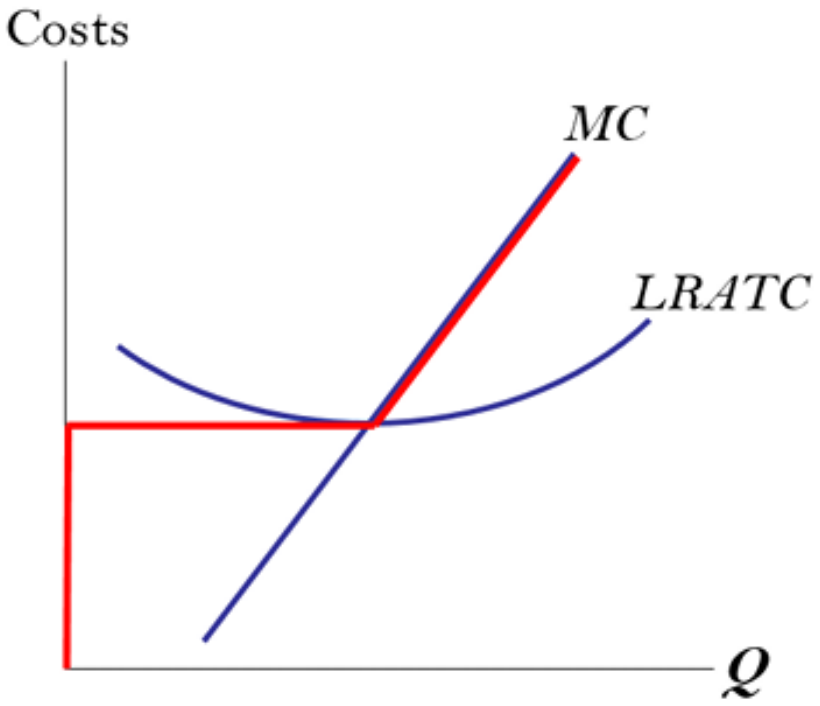
Competitive Markets

- **Shut down (TR < VC, or P < AVC)**



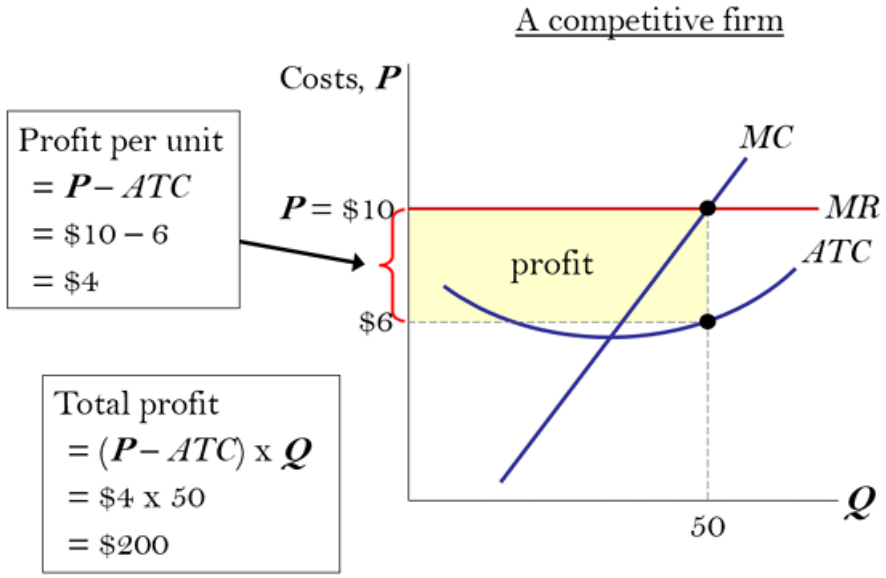
Competitive Markets

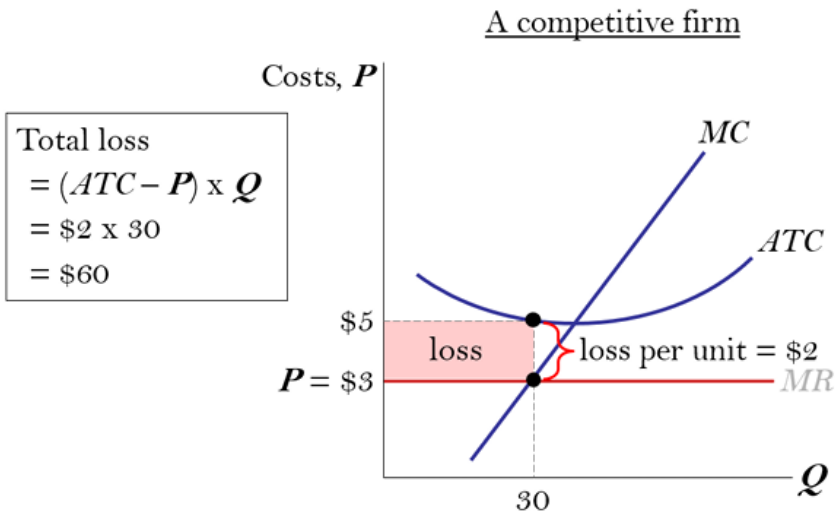
- **Exit (TR < TC, or P<ATC)**



Competitive Markets

Total profit/loss





P> ATC→ not shut down

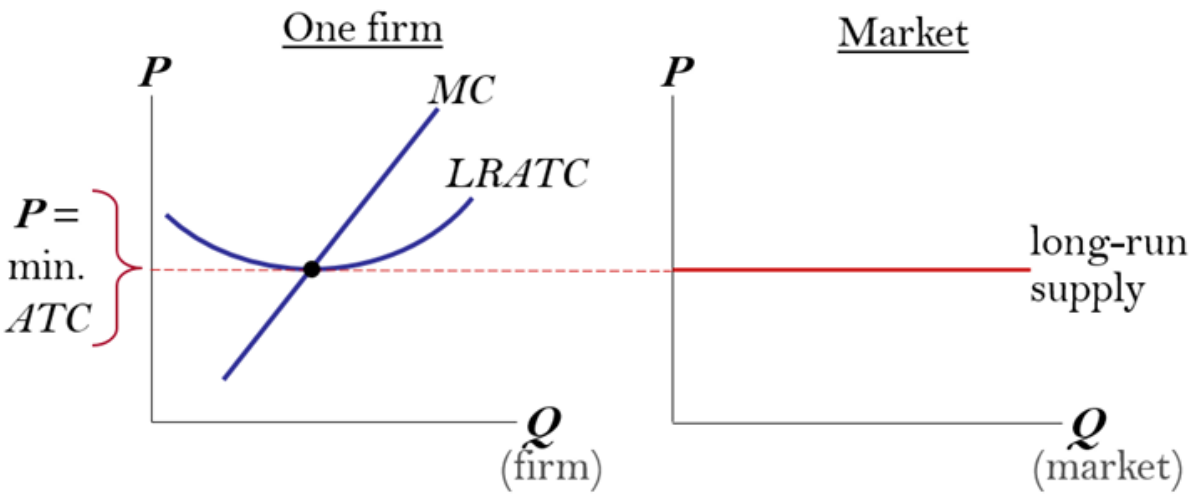
When AVC<3 → not shut down

Competitive Markets

- Zero-profit Condition (Economic profit = 0; Accounting profit>0)
  - P = ATC
  - P = MR = MC
  - P = min ATC

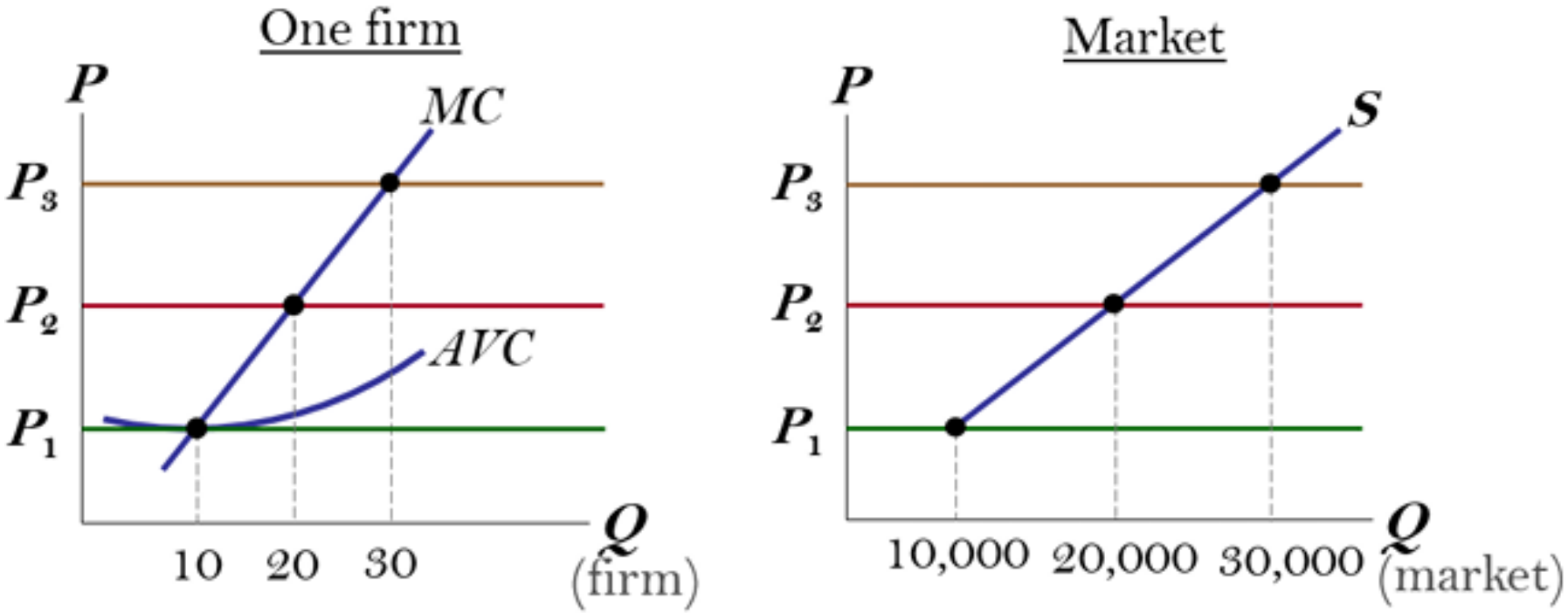
In the long run, the typical firm earns zero profit.

The LR market supply curve is horizontal at *P* = minimum *ATC*.



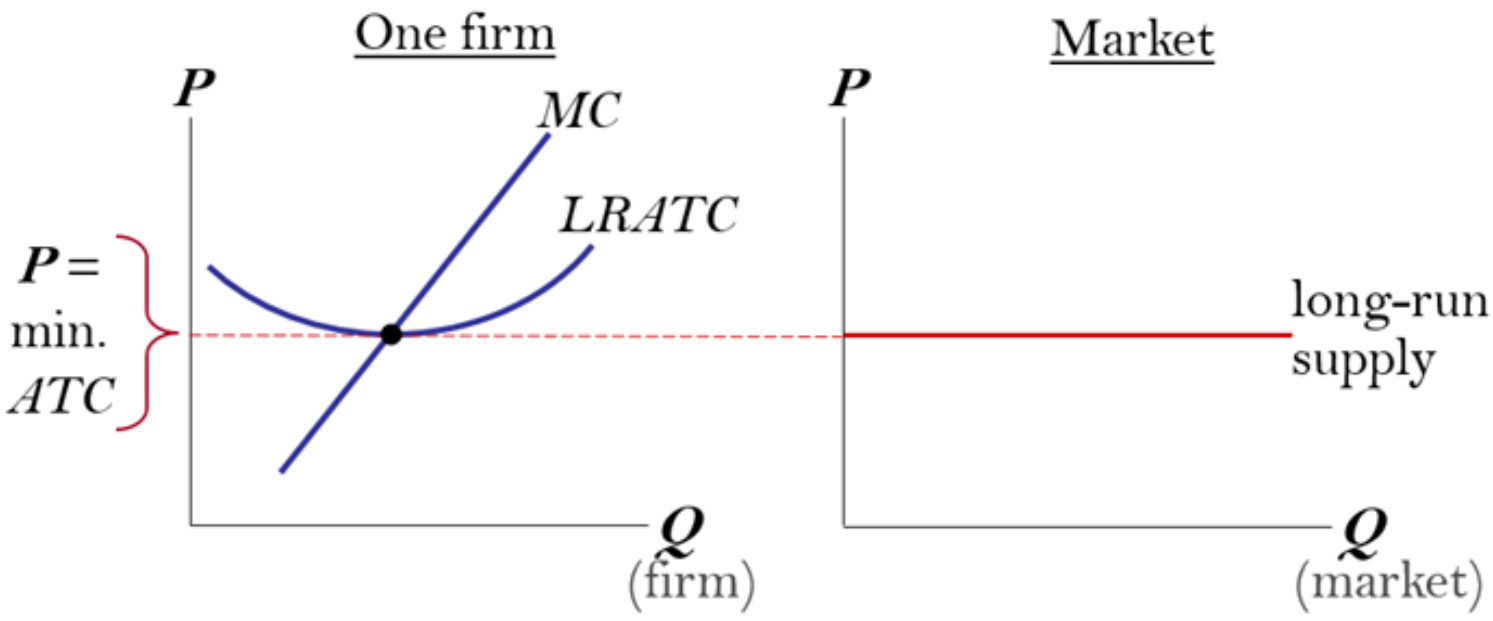
Competitive Markets

- SR Supply Curve



Competitive Markets

- LR Supply Curve



Production and Cost

- Question -SR

Given: $TC_{SR} = f(q)$ C $Q^D = D(P)$ N

Then: $MC(q) = \frac{df(q)}{dq}$ $AVC(q) = \frac{f(q) - C}{q}$

$$q^s_{SR}(P) = \begin{cases} MC^{-1}(P) & \text{if } P \geq \min(AVC) \\ 0 & \text{otherwise} \end{cases}$$

$$Q^s_{SR}(P) = N \cdot q^s_{SR}(P)$$

$$D(P) = Q^s_{SR}(P)$$

$$\pi = P^* \cdot q^s_{SR}(P^*) - f\left(q^s_{SR}(P^*)\right)$$

Production and Cost

- Question -LR

Given: $TC_{LR} = g(q)$ $\quad Q^D = D(P)$

Then: $LMC(q) = \dfrac{dg(q)}{dq}$ $\quad LAC(q) = \dfrac{g(q)}{q}$

$$P^* = \min(LAC)$$

$$q^* = \arg\min LAC \qquad Q^* = D(P^*)$$

$$N^* = \frac{Q^*}{q^*}$$

$$\pi = P^* \cdot q^* - g(q^*) = 0$$

Production and Cost

- Example: N = 40 $TC = q^2 + 8, \quad Q^D = 120 - 10P,$

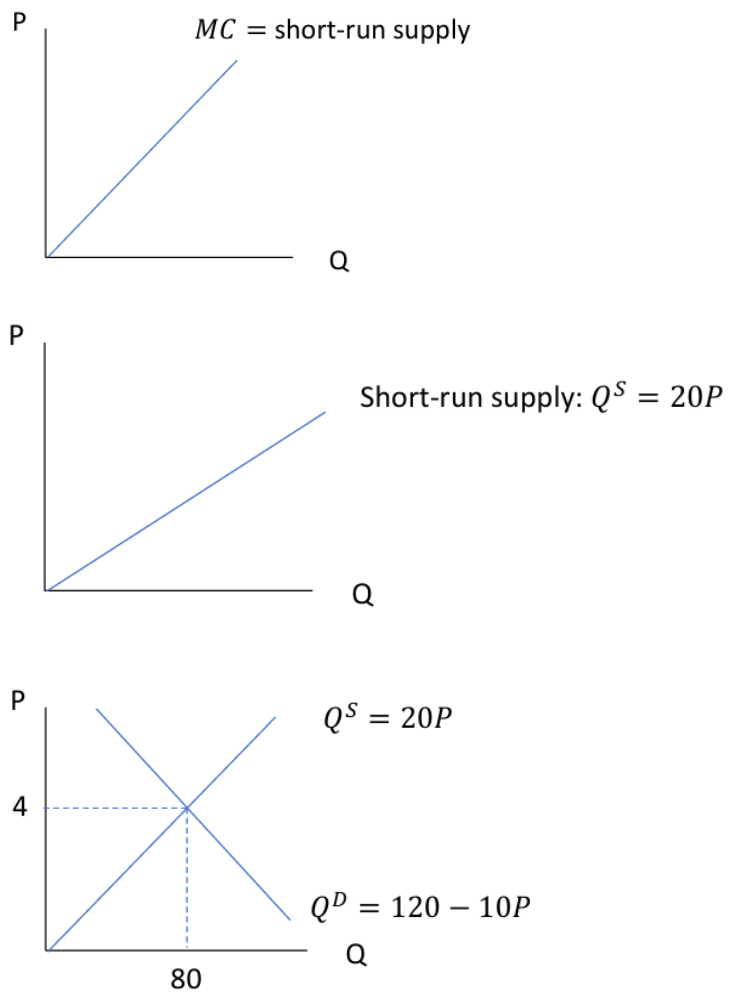
Individual: $MC = \frac{\delta TC}{\delta q} = 2q = \mathrm{p} \rightarrow \mathrm{q} = \frac{\mathrm{p}}{2}$

$$AVC = \frac{q^2}{q} = q \qquad ATC = \frac{TC}{q} = q + \frac{8}{q}$$

Market: $Q^S = \frac{P}{2} * 40 = 20P$

Equilibrium: $Q^D = Q^S$

Profit: $\Pi = \mathrm{q}^*$ (P-ATC)



Production and Cost

- Example: $TC = q^2 + 8$, $Q^D = 120 - 10P$

Individual: $MC = \frac{\delta TC}{\delta q} = 2q$ $\quad ATC = \frac{TC}{q} = q + \frac{4}{q}$

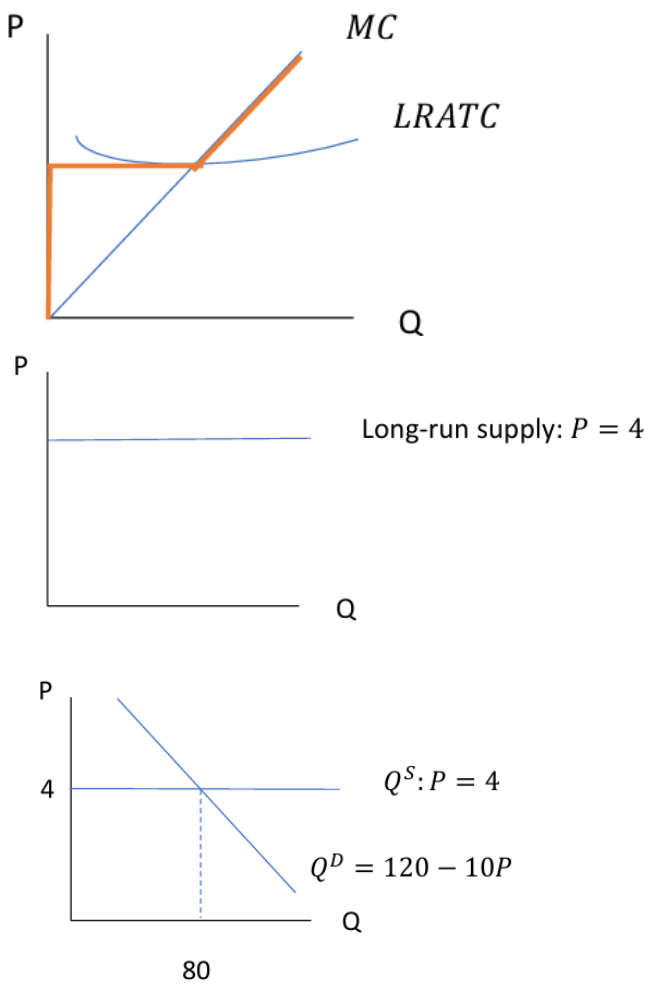
Market: horizontal at the level where price is equal to the minimum ATC (intersection with MC)

$$P = \min ATC \qquad q + \frac{4}{q} = 2q \qquad q = 2, P = LRATC = 4$$

Equilibrium: $Q^D = 120 - 10P$ $\quad P = 4, Q^D = 80$

Number of firms: $\frac{80}{2} = 40$

Profit: 0



Competitive Markets

- LR Supply Curve
  - Can be upward-sloping when
    - Firms have different costs
    - Or costs rise as firms enter the market

Monopoly

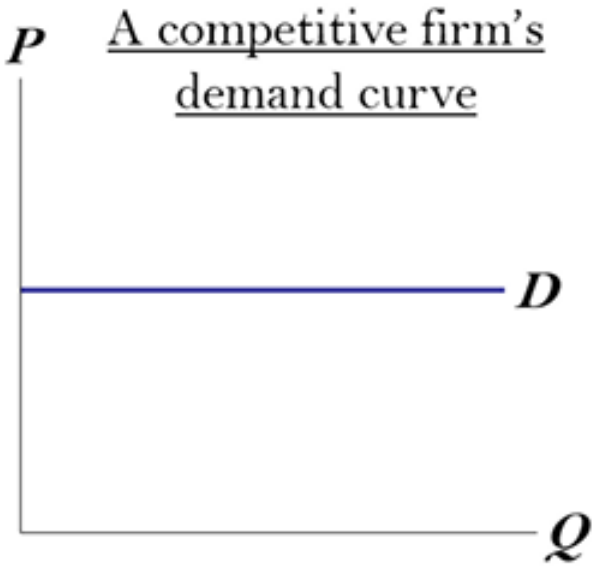
- Definitions and three barriers
- Demand curve
- MR, Profit-Maximization, Profit
- A Rule of Thumb for Pricing
  - Markup

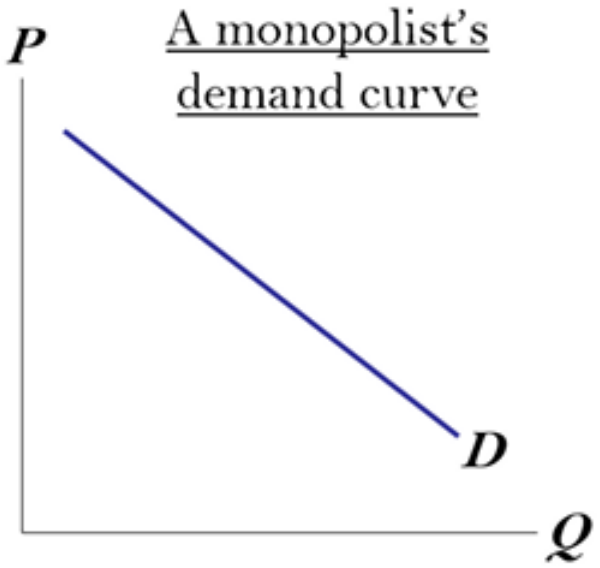
Monopoly

- Monopoly (has market power)
  - A firm that is the sole seller of a product without close substitutes
  - Three Barriers to Entry
    - Monopoly resources
    - Government regulation
    - Fill in the production process

Monopoly

- Demand Curves





Market demand curve

for a competitive market

→ downward

Monopoly

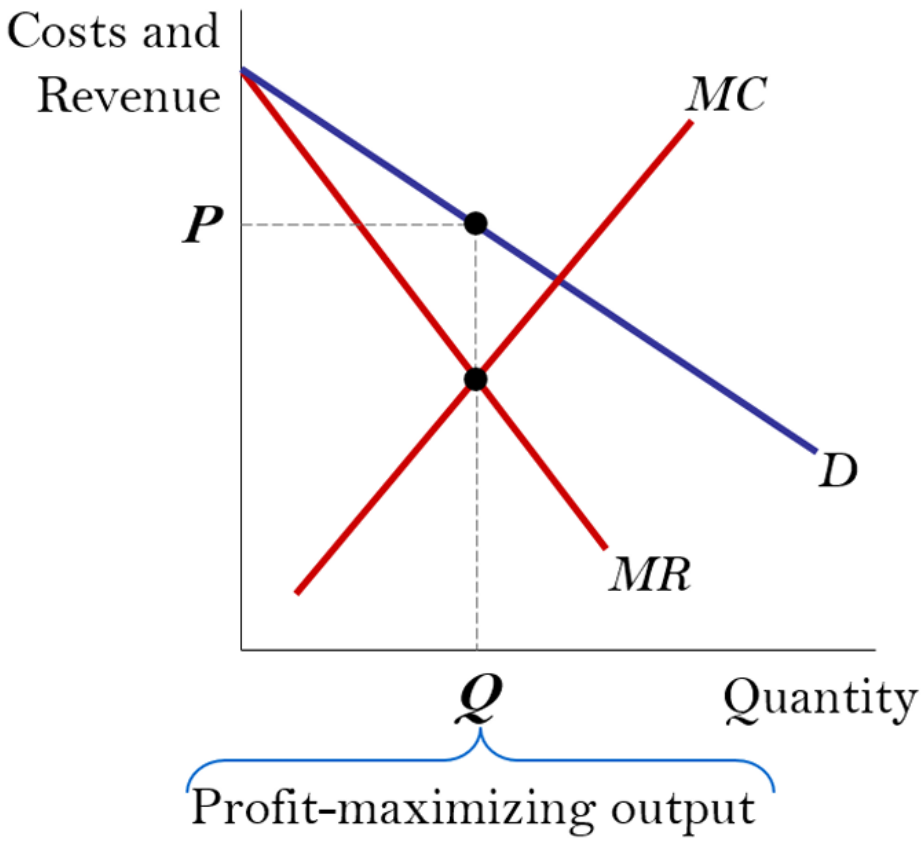
- **P = AR**
  - AR = TR/Q = P*Q/Q = P holds for all the firms

Monopoly

- **MR < P**
  - To sell an additional unit of output, the monopolist **must lower the price for all units sold**, not just the new one.
    - Selling an extra unit increases revenue by the unit price (+P)
    - Lowering the price reduces revenue from all previous units sold ($-Q \times \Delta P$)
  - MR = Price of the new unit (P)- Revenue loss from lowering price

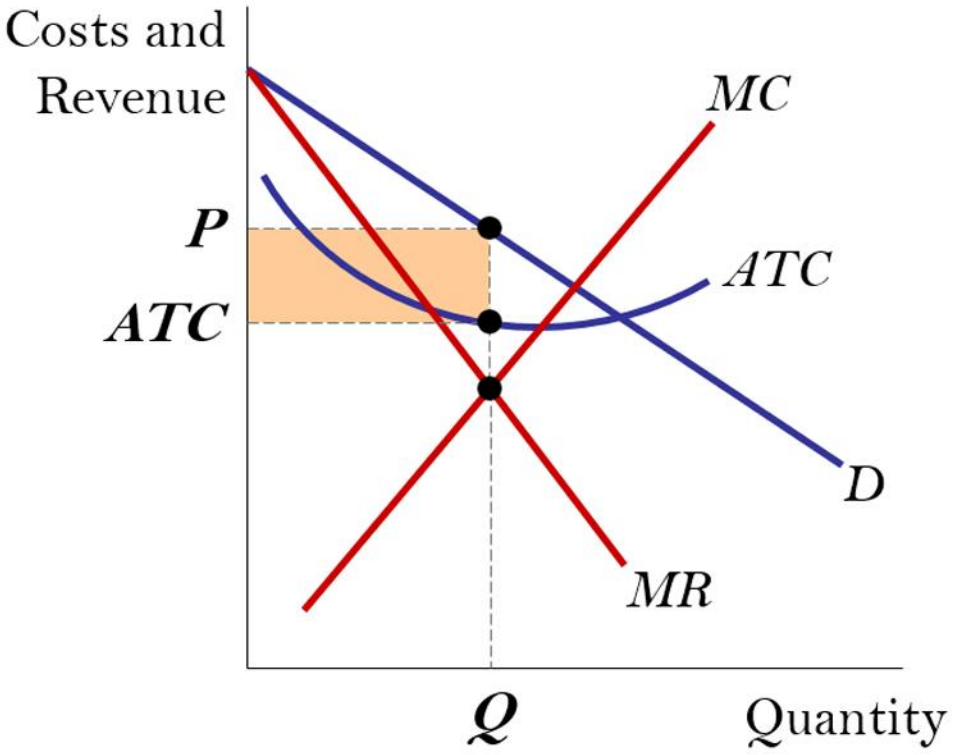
Monopoly

- **Profit Maximization**
  - **MR = MC**



Monopoly

- **Profit Maximization**
  - **MR = MC**



Monopoly

Dimensions	Competitive Markets	Monopoly
Market	Infinite firms without market power	Single firm with market power
Product	Identical	No substitute
Price	Price taker	Price maker
MR	P = MR	P > MR
AR	P = AR	P = AR
Profit Maximization	MR = MC (P = MC = min ATC)	MR = MC (P > MC, P > ATC)
Social Welfare	Maximized	Deadweight Loss

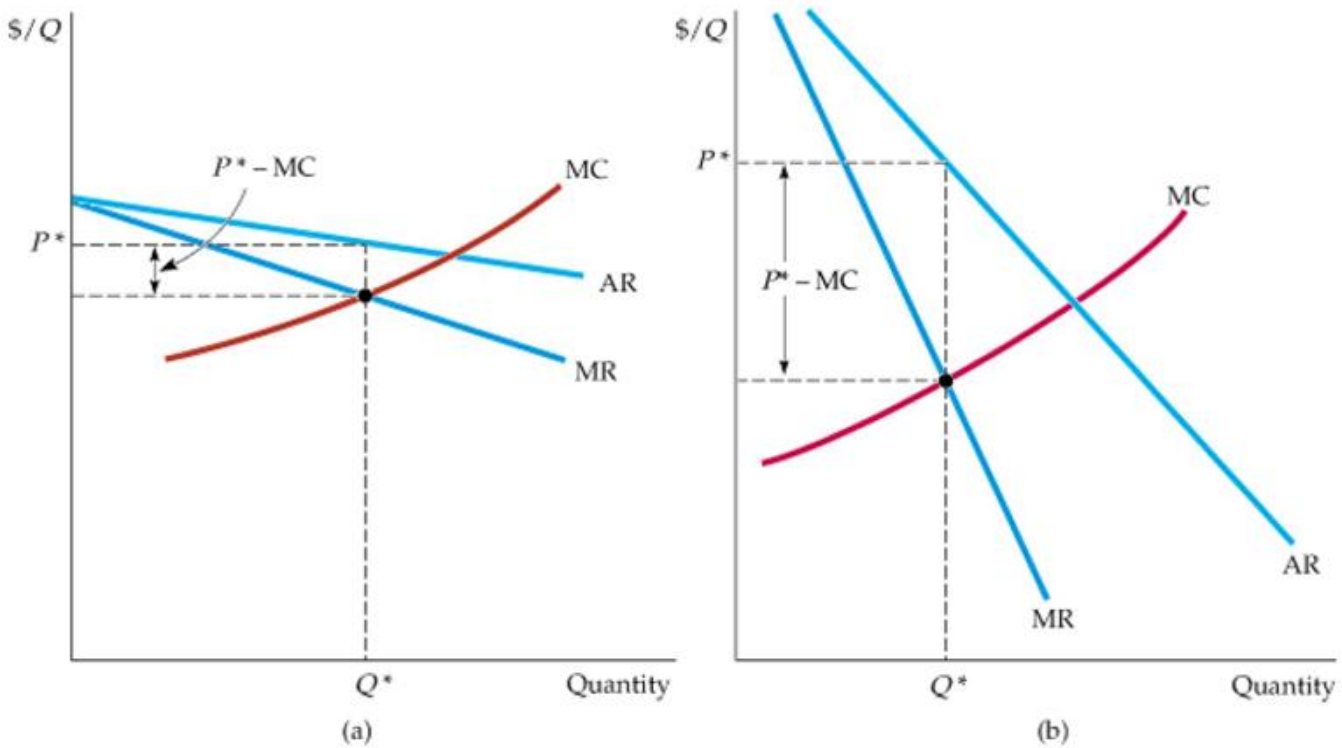
Monopoly

- A Rule of Thumb for Pricing

Markup:

$$\frac{P - \mathrm{MC}}{P} = -\frac{1}{E_d} \qquad \frac{P - MC}{P} = \frac{1}{|E_d|}$$

$$P = \frac{\mathrm{MC}}{1 + (1/E_d)} \qquad P = \frac{MC}{1 - \frac{1}{|E_d|}}$$

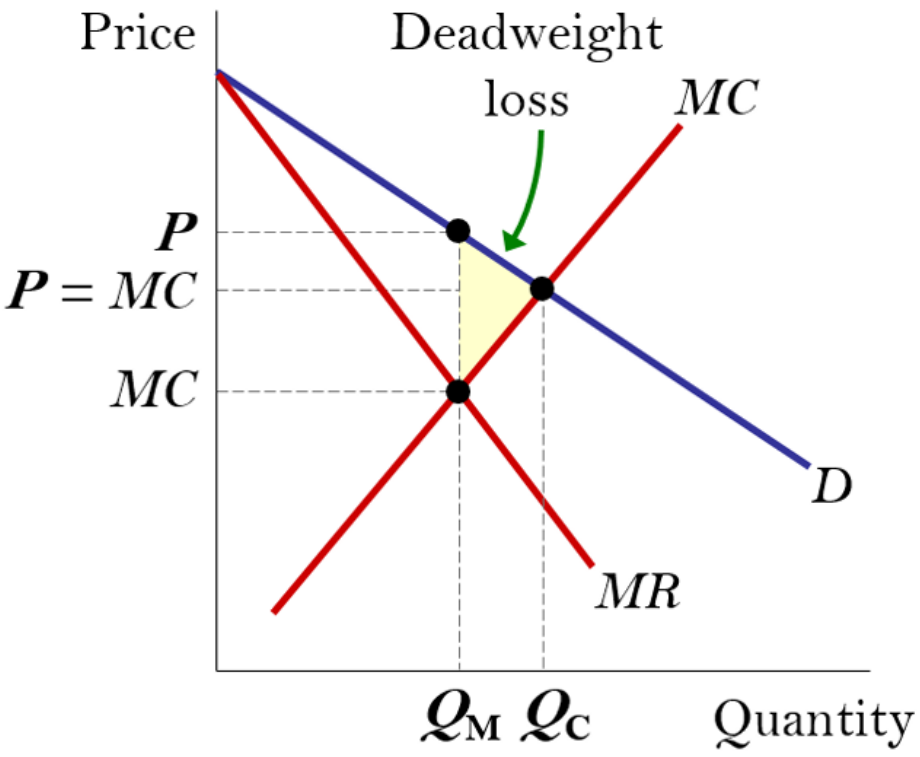


Elastic → markup is small; little monopoly power

Inelastic → markup is big; more monopoly power

Monopoly

- Welfare Cost



Thanks!

Good Luck!